



(12) **United States Patent**
Ouimet et al.

(10) **Patent No.:** **US 12,411,890 B2**
(45) **Date of Patent:** **Sep. 9, 2025**

(54) **SYSTEM TO CORRELATE VIDEO DATA
AND CONTEXTUAL DATA**

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(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 189 days.

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(22) Filed: **Jan. 28, 2016**

(Continued)

(65) **Prior Publication Data**

US 2017/0161382 A1 Jun. 8, 2017

Primary Examiner — William P Bartlett

Related U.S. Application Data

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(60) Provisional application No. 62/264,549, filed on Dec. 8, 2015.

(57) **ABSTRACT**

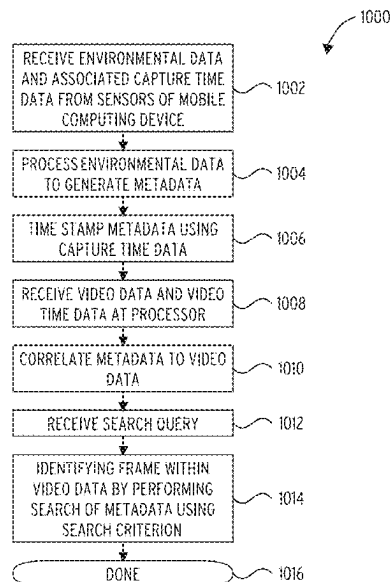
(51) **Int. Cl.**
G06F 16/78 (2019.01)
G06F 3/16 (2006.01)
(Continued)

In some embodiments, a method of processing image data may include receiving environmental data and associated capture time data from a sensor of a mobile computing device, the capture time data reflecting capture time of the environmental data; processing the environmental data to generate metadata; time stamping the metadata using the capture time data; receiving video data and video time data at a processor; correlating the metadata to the video data using the capture time data and the video time data; receiving a search query; and/or identifying a frame within the video data by performing a search of the metadata using the search criterion.

(52) **U.S. Cl.**
CPC **G06F 16/78** (2019.01); **G06F 16/434** (2019.01); **G06F 16/487** (2019.01);
(Continued)

(58) **Field of Classification Search**
CPC .. G06F 17/30817; G06F 16/78; G06F 16/487;
G06F 16/489; G06F 16/434;
(Continued)

16 Claims, 17 Drawing Sheets



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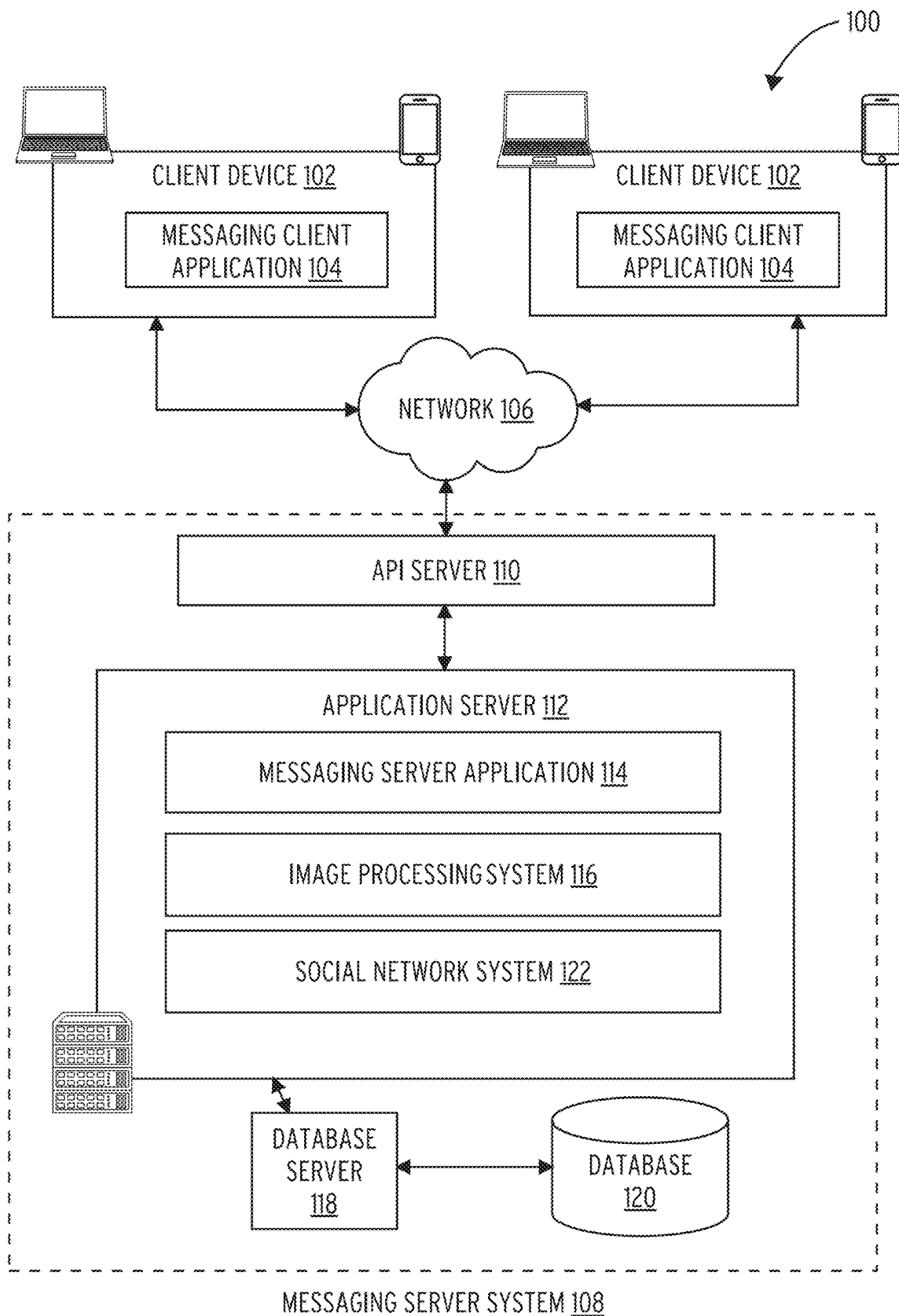


FIG. 1

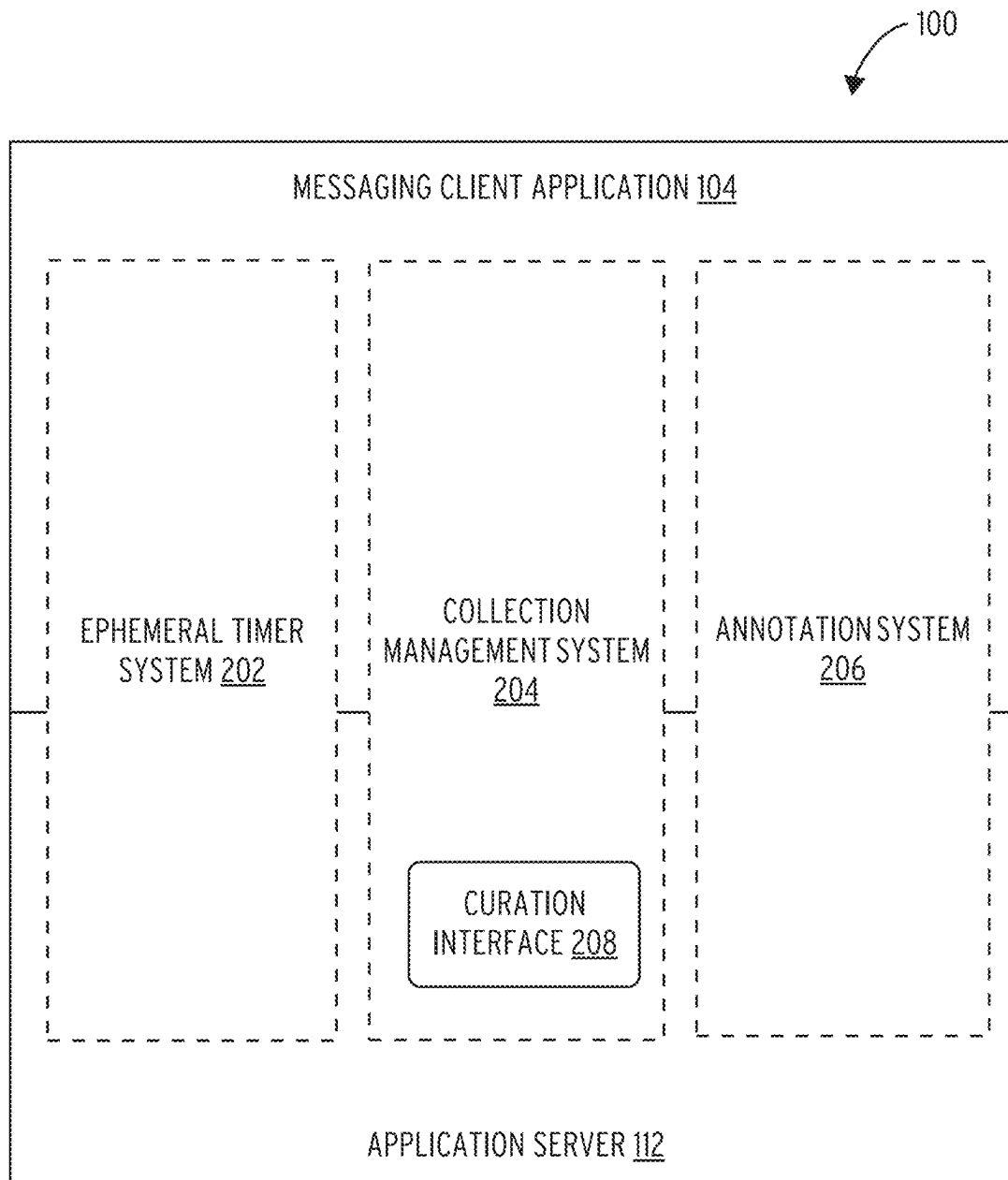


FIG. 2

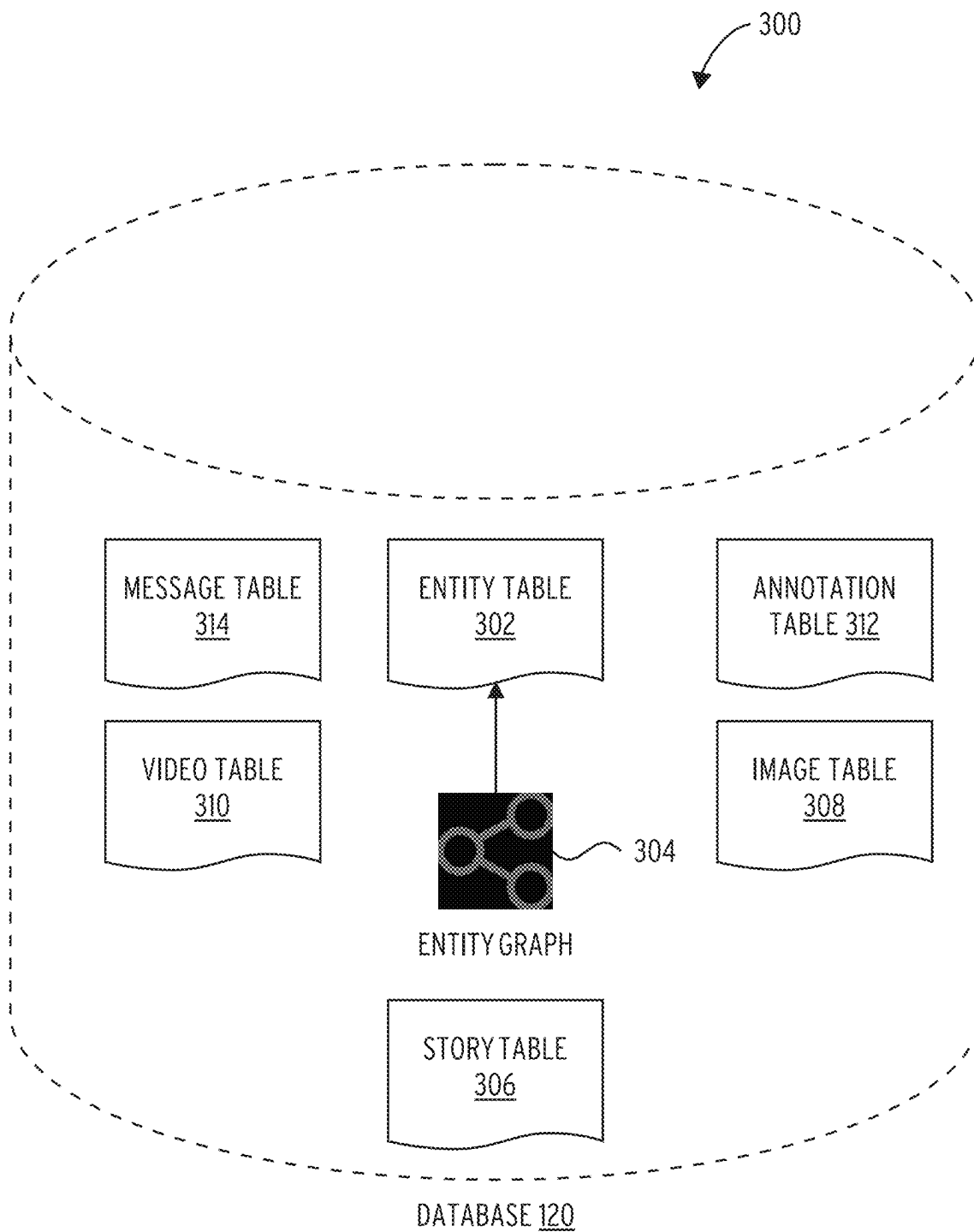


FIG. 3

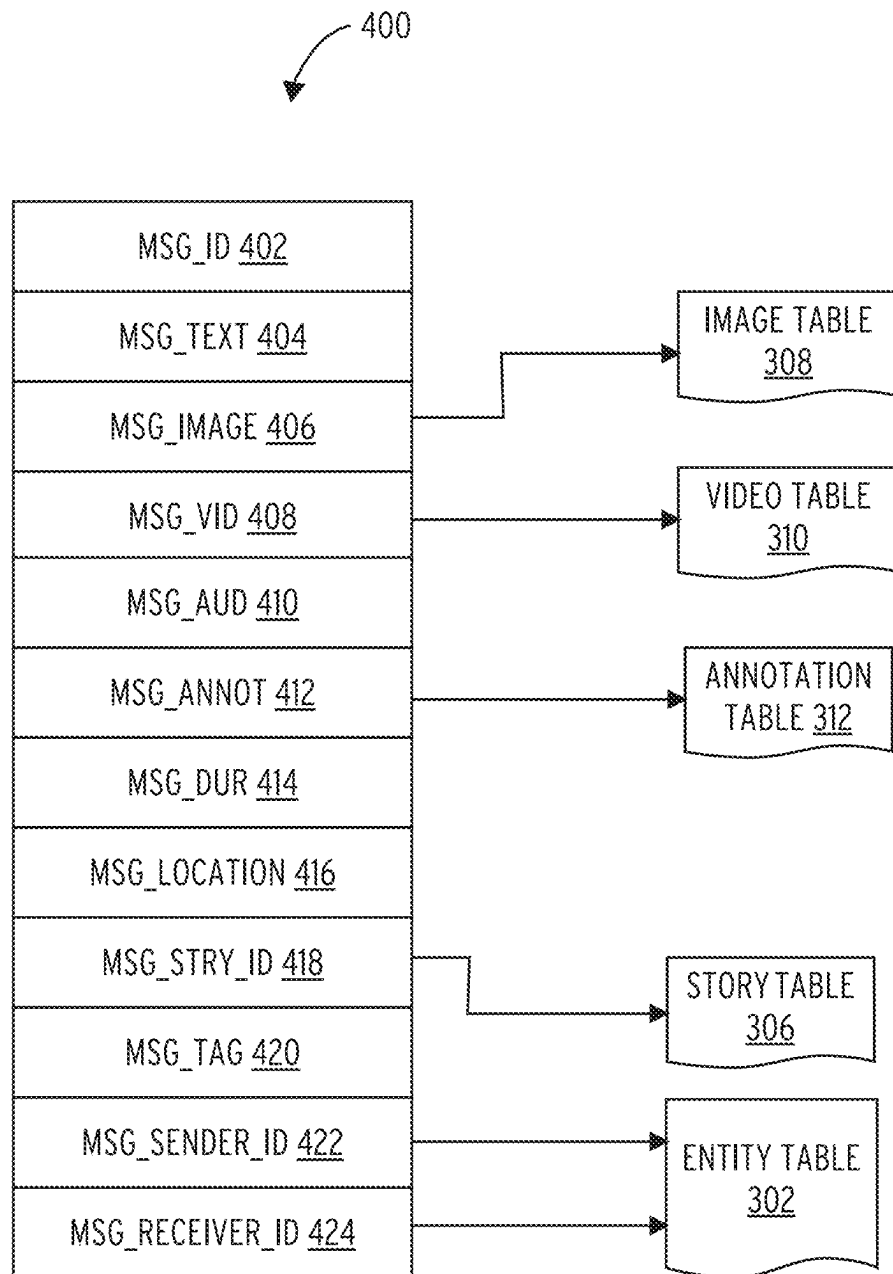


FIG. 4

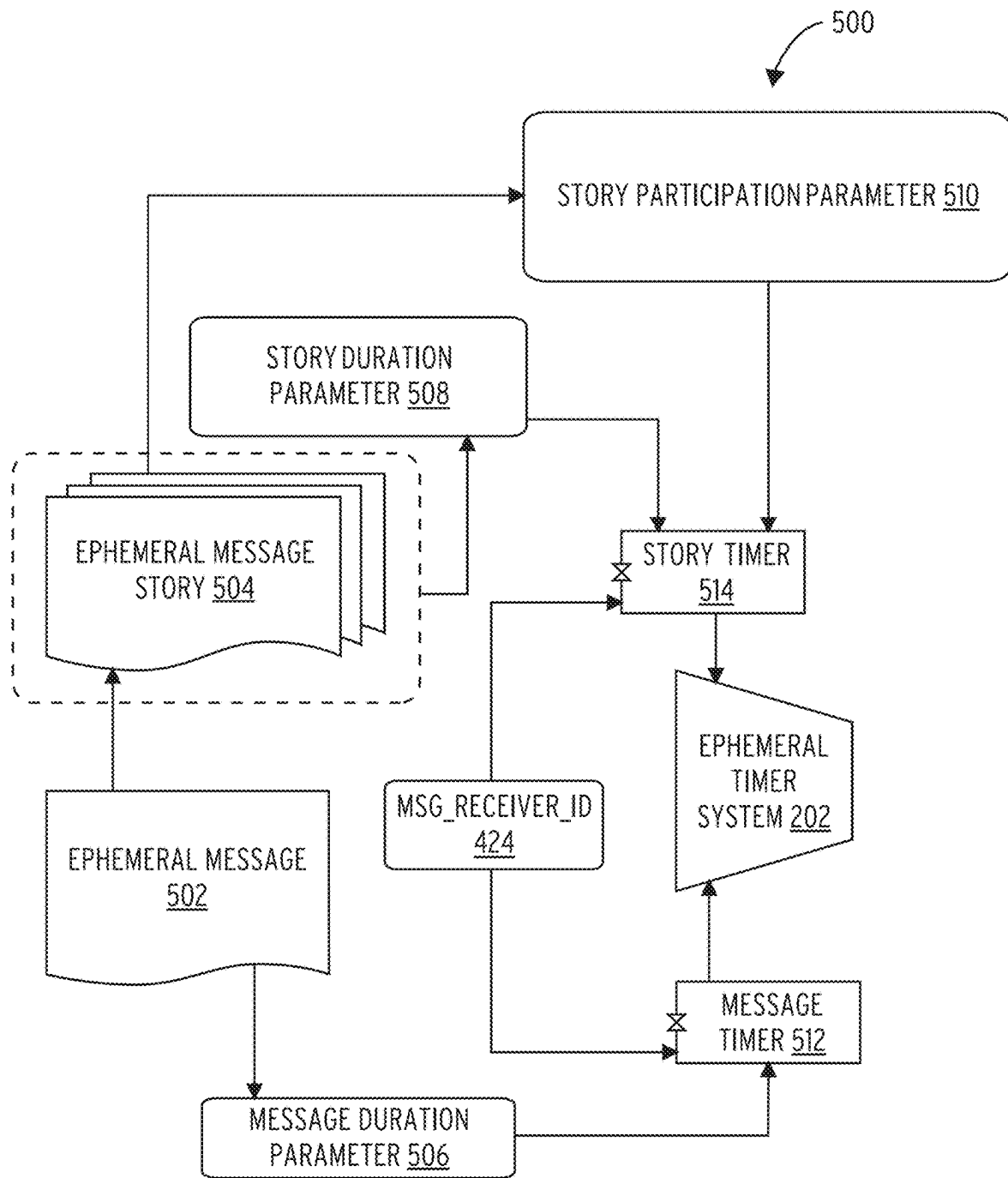


FIG. 5

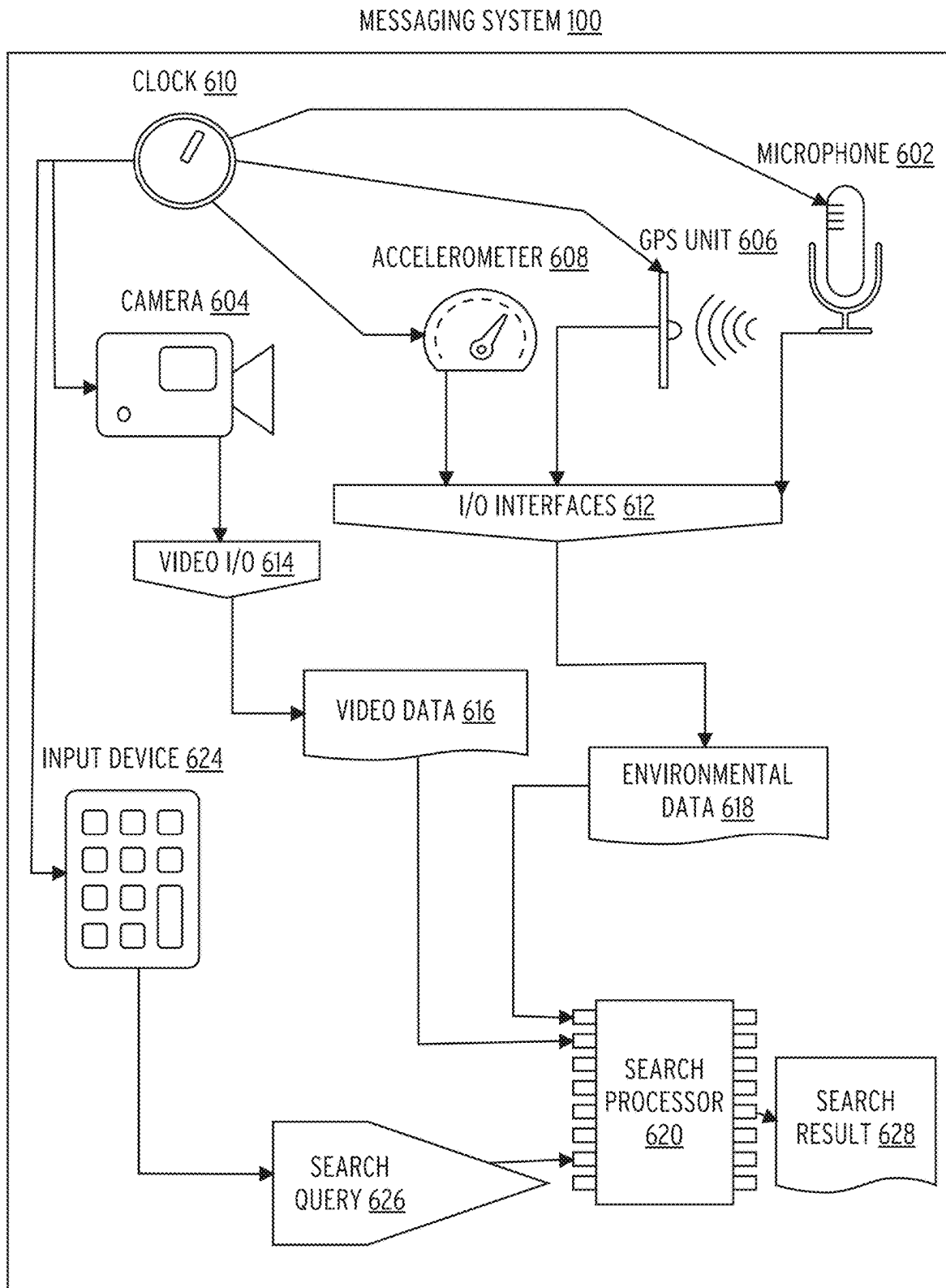


FIG. 6

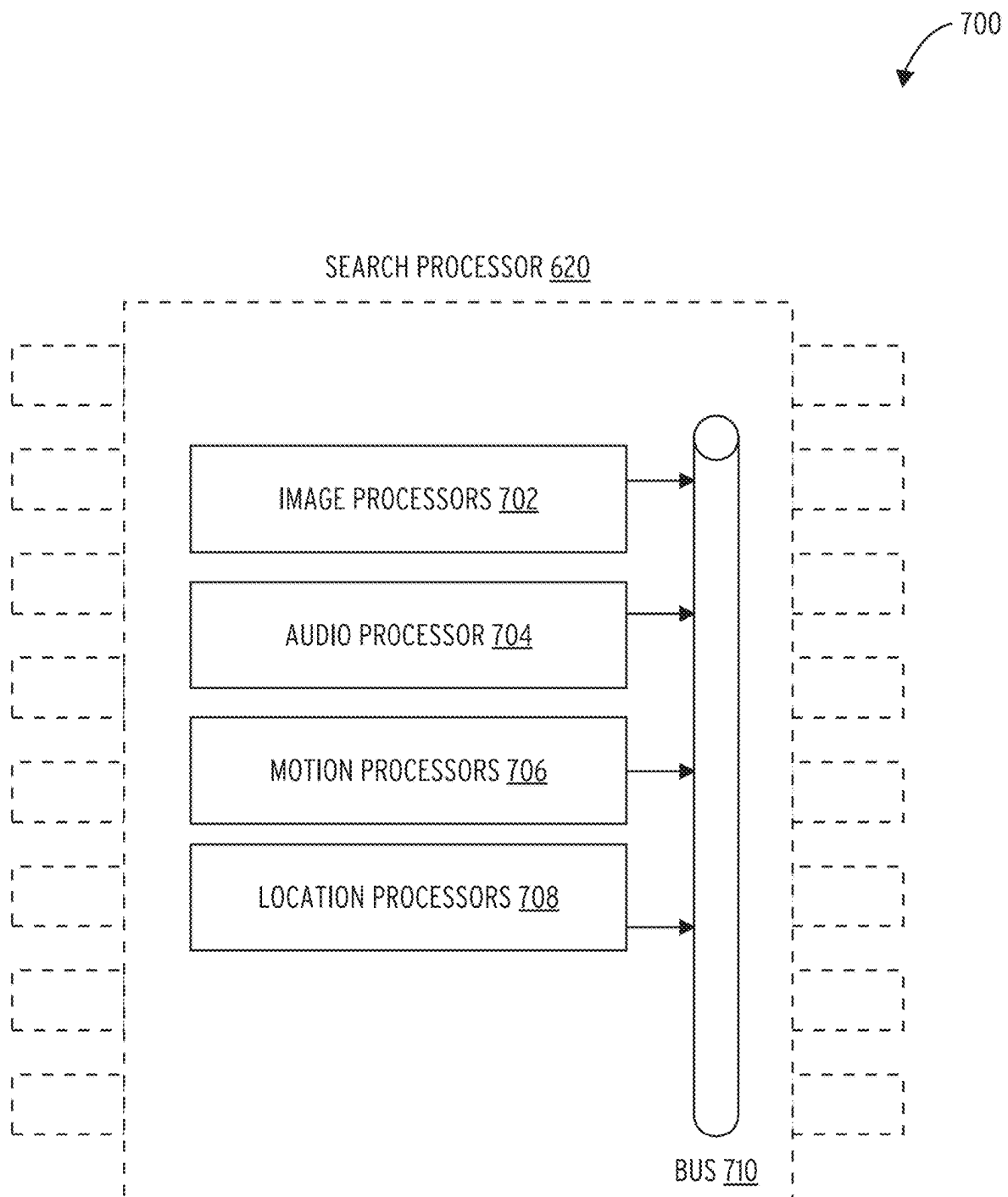


FIG. 7

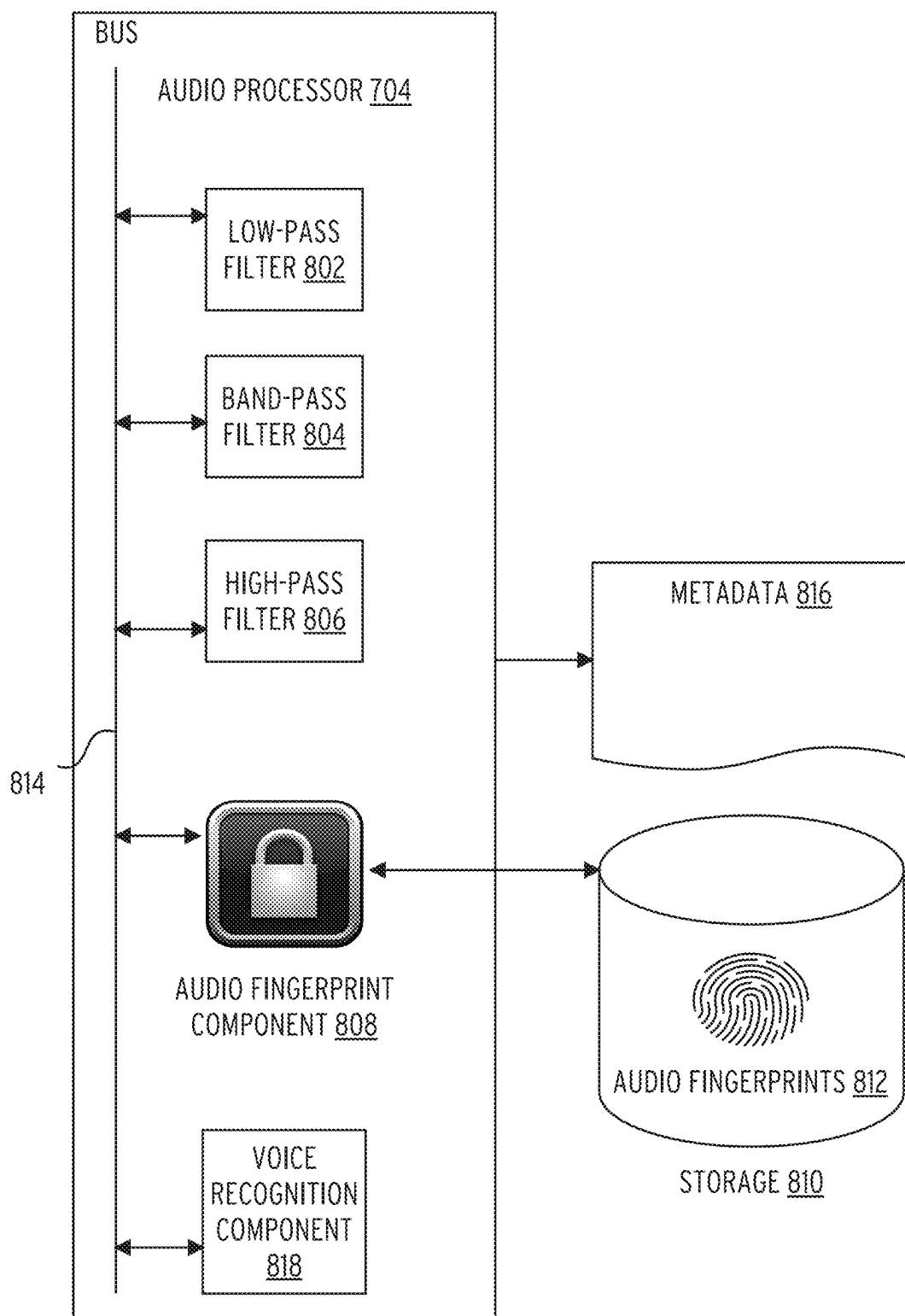


FIG. 8

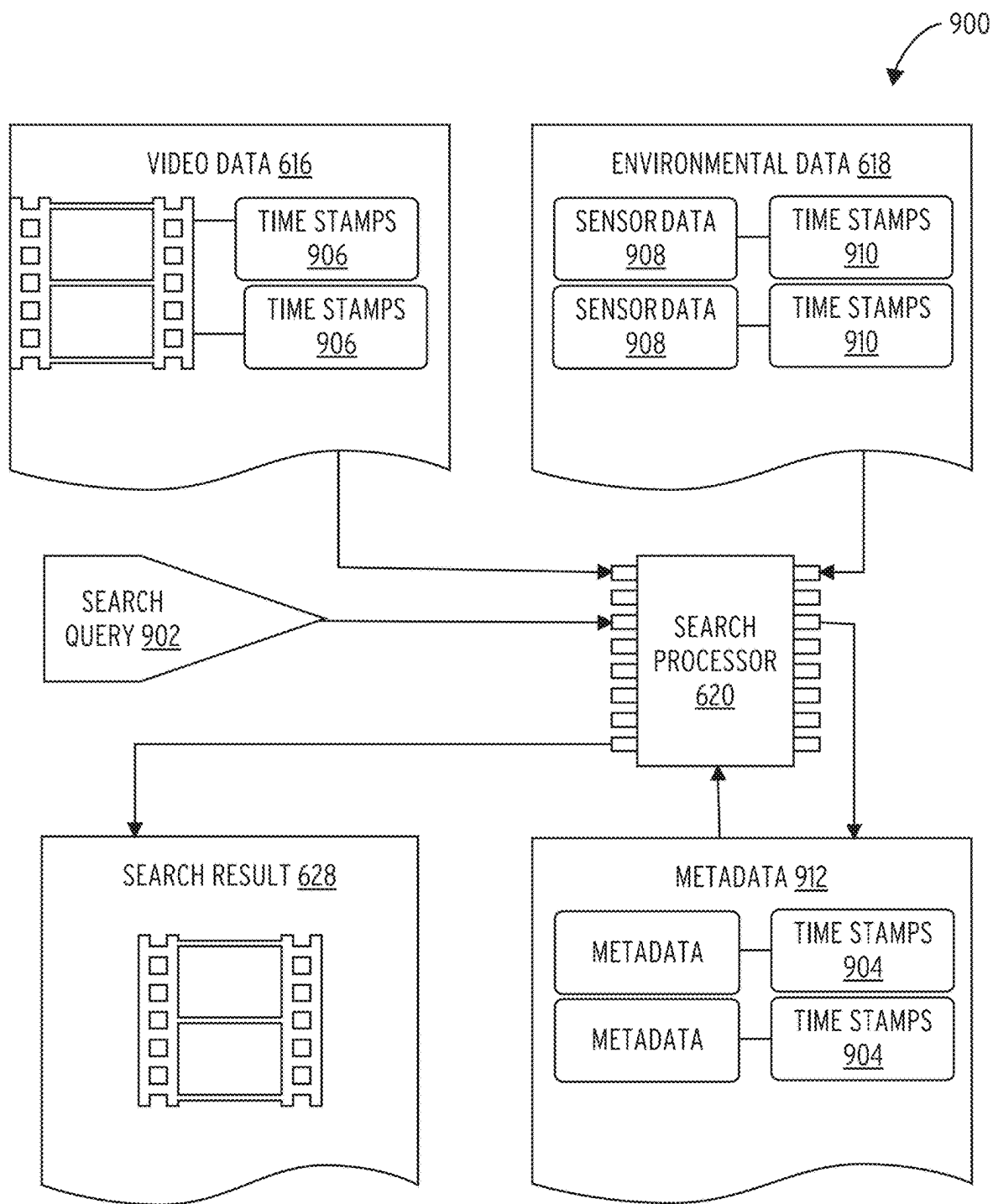


FIG. 9

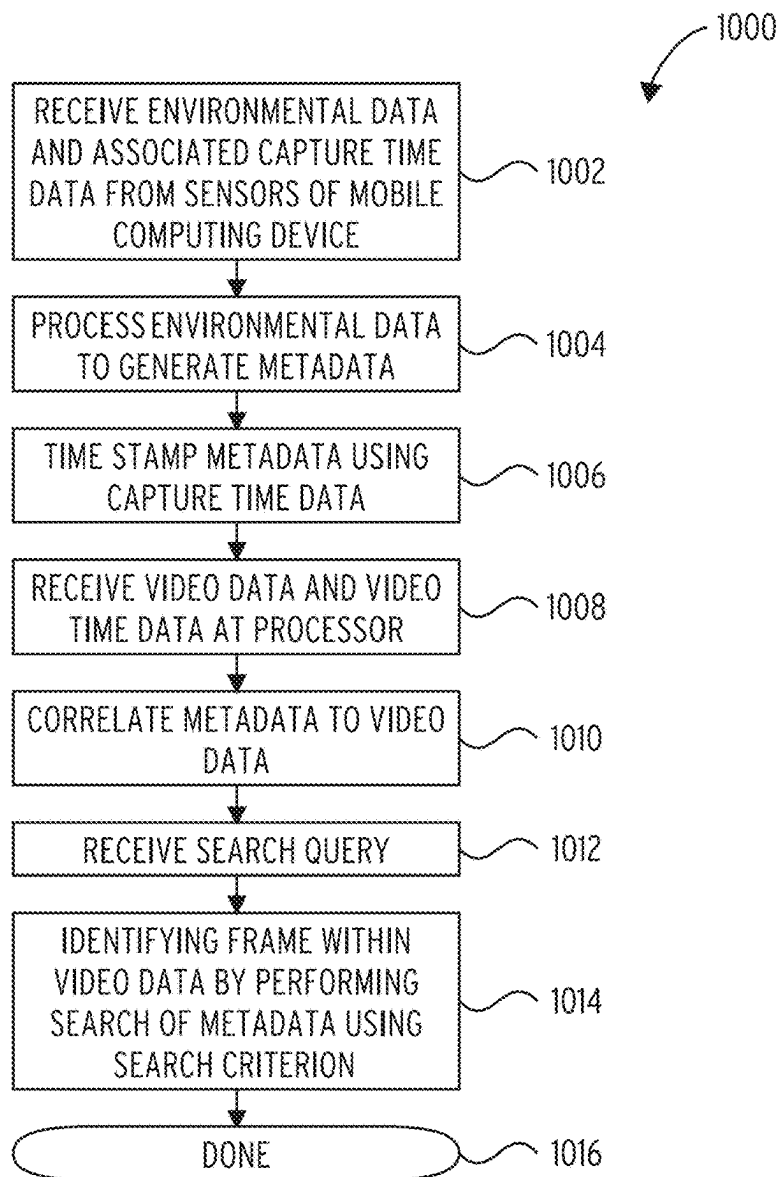


FIG. 10

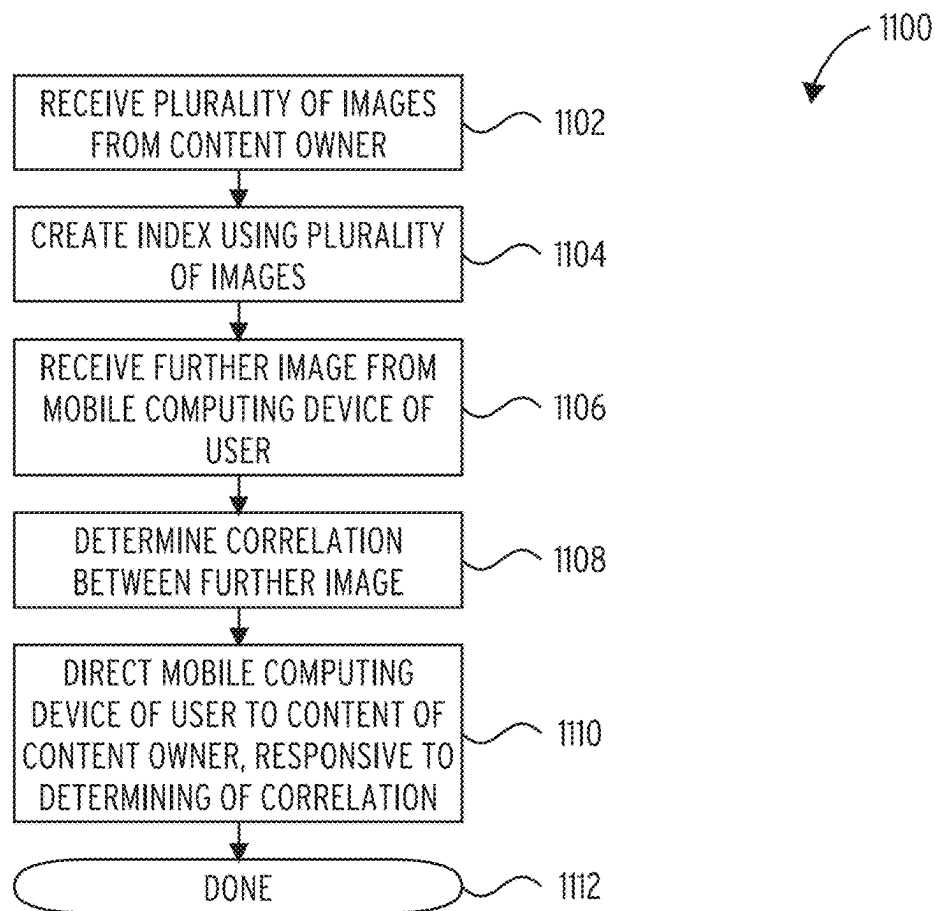


FIG. 11

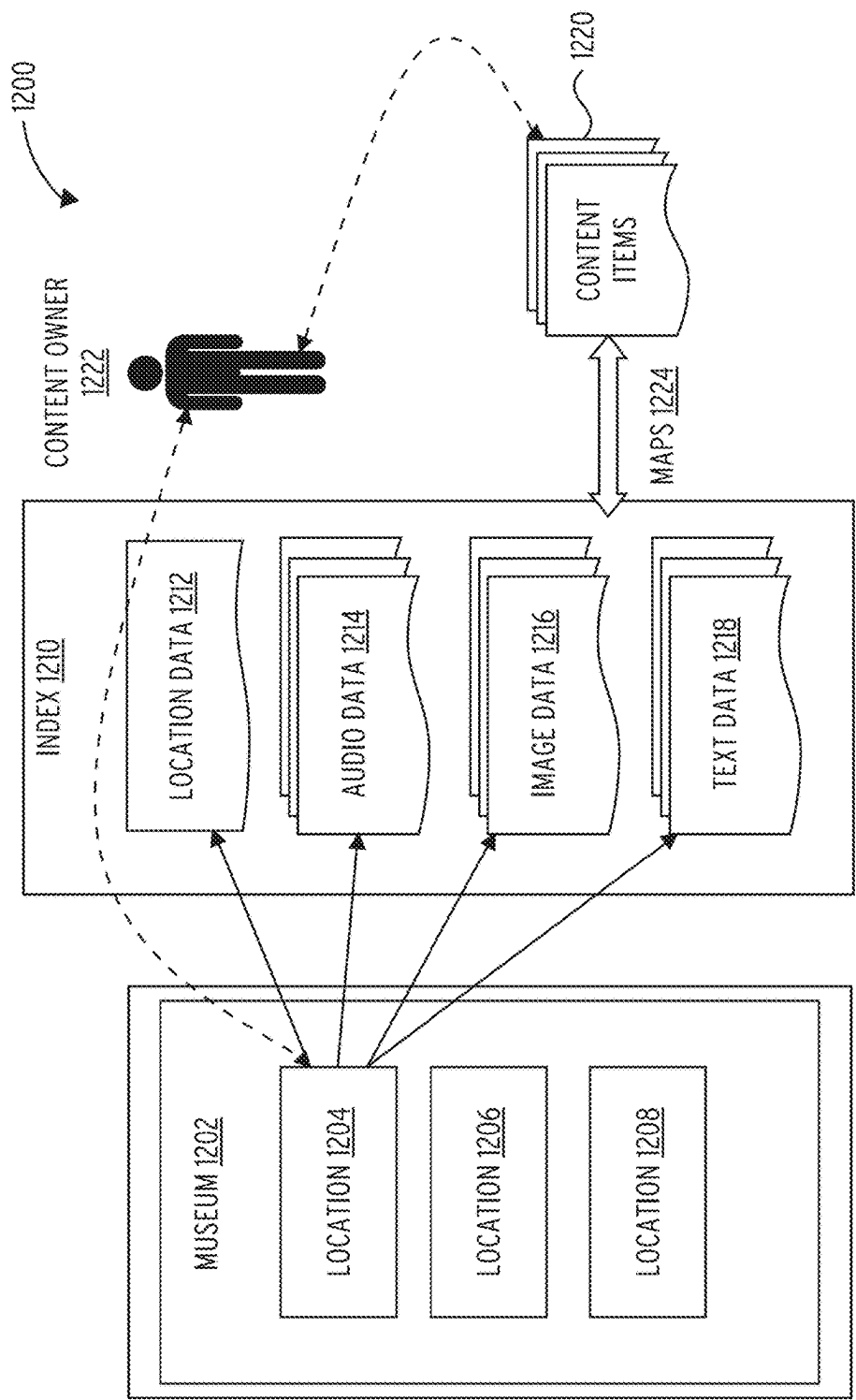


FIG. 12

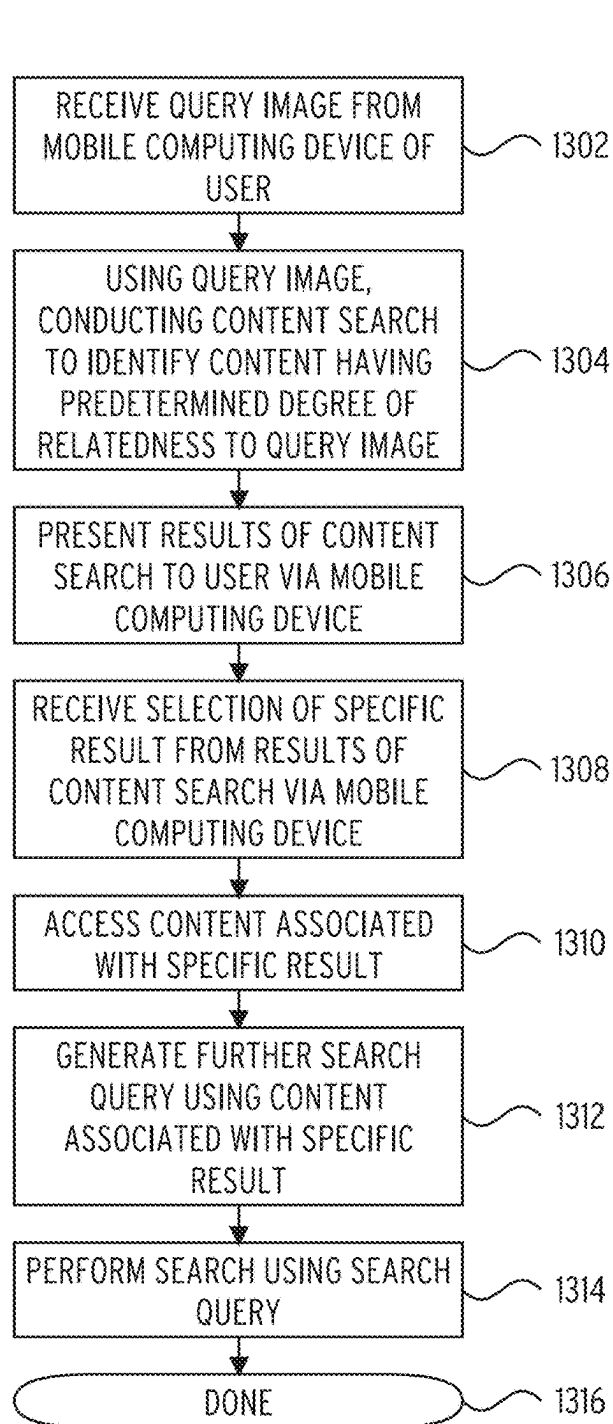


FIG. 13

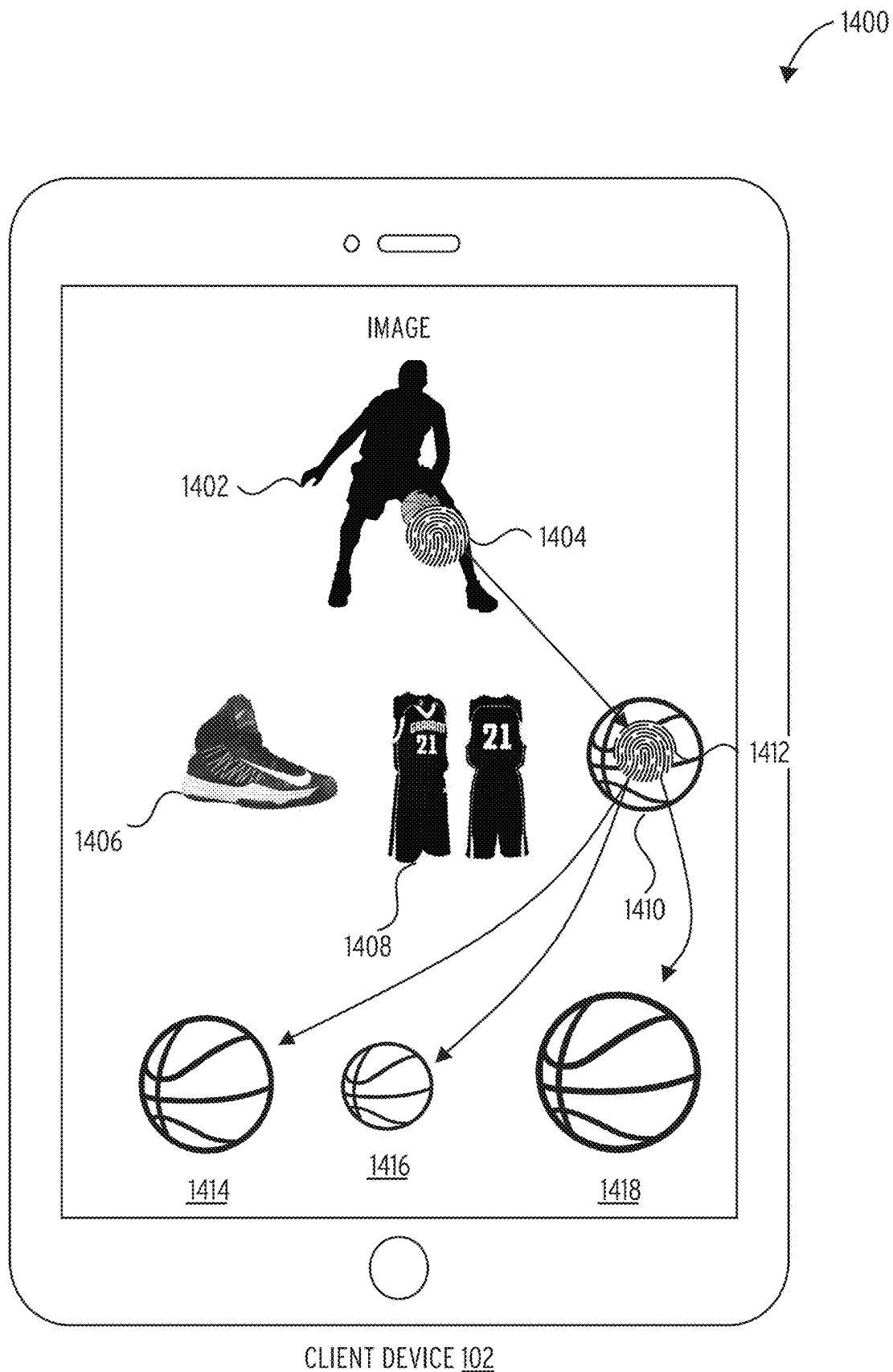


FIG. 14

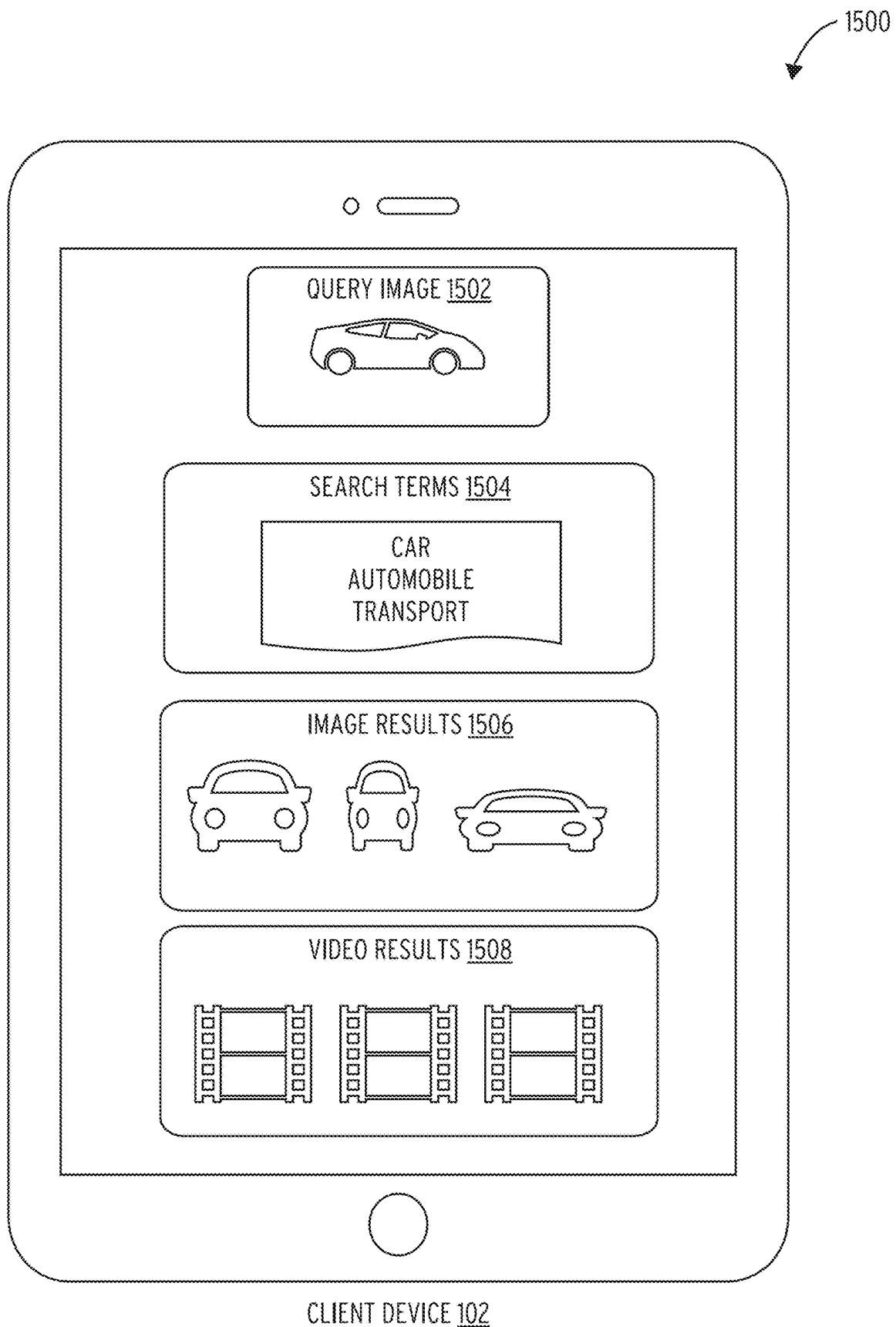


FIG. 15

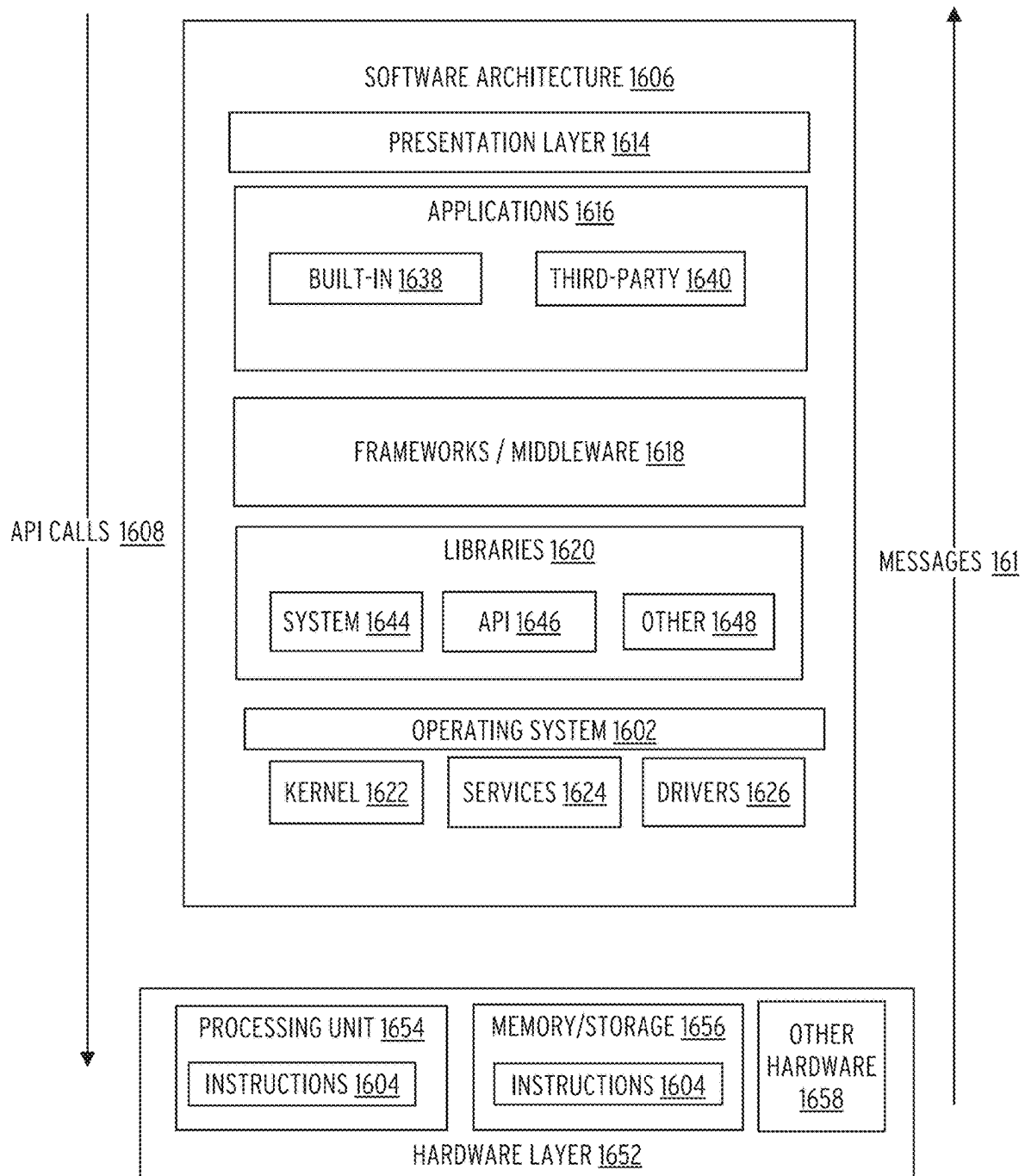


FIG. 16

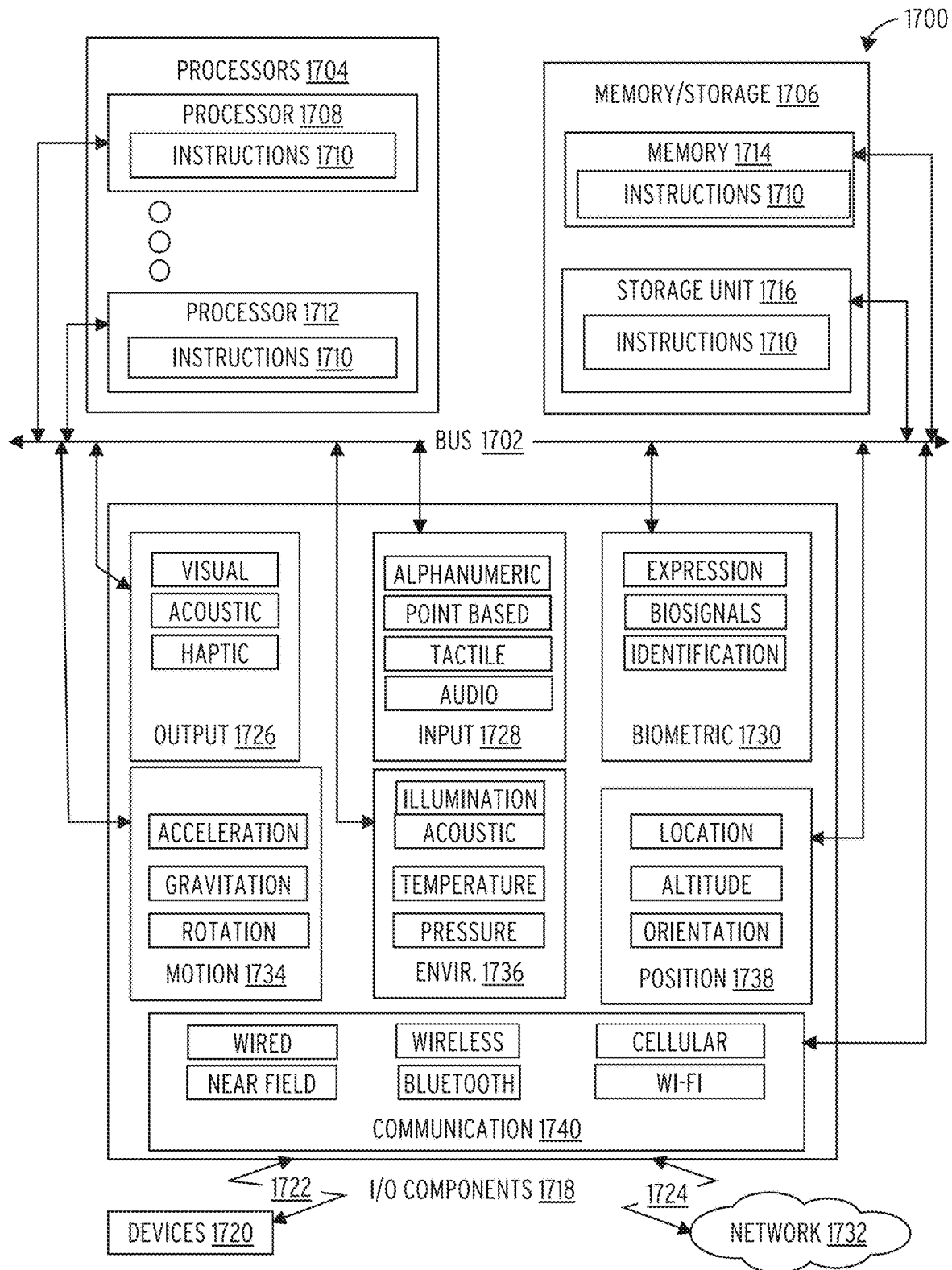


FIG. 17

1

SYSTEM TO CORRELATE VIDEO DATA
AND CONTEXTUAL DATACROSS-REFERENCE TO RELATED
APPLICATIONS

This application claims the benefit of priority of U.S. Provisional Application No. 62/264,549, filed Dec. 8, 2015, which is hereby incorporated by reference in its entirety.

BACKGROUND

Image-based searching has become more prevalent, and of greater interest, with the increased adoption of mobile technologies, and particularly the widespread use of mobile devices having integrated camera systems. The ubiquitous availability of mobile devices and cameras as presented the opportunity for both hardware and software companies to provide image-based searching capabilities on these mobile devices to their users.

Image-based searching typically requires the user to specify a query on the basis of an image, rather than text. This presents a number of technical challenges, particularly with respect to user interface technologies and paradigms. A number of prior art technologies to improve upon image of the messaging have been proposed.

US patent application publication number 2015/0052431 describes touch controls for image-based search. Specifically, gesture information, received via a touch-sensitive screen, is used to generate a selection area on an image. A processor circuit extracts an image portion corresponding to the selection area, and performs an image-based search using the extracted image portion.

US patent application publication number 2014/0188894 describes receiving gesture data based on a user gesture interacting with a portion of displayed content. A subset of the displayed content is identified based on the gesture data, and a set of candidate queries is identified based on the subset of content.

US patent application publication number 2012/016 6472 describes a system for developing a query by manipulating tangible (physical) objects that represent the query.

U.S. Pat. No. 8,850,490 describes an Internet-based content platform that includes a search graphical user interface to present search results in response to a search query from an end user. The search can be a text-based search, an image-based search, and audio-based search, a video-based search, a biometric input-based search et cetera.

While the above described technologies have advanced image-based search technologies in general, a number of technical challenges remain.

BRIEF SUMMARY

In some embodiments, a method of processing image data may include receiving environmental data and associated capture time data from a sensor of a mobile computing device, the capture time data reflecting capture time of the environmental data; processing the environmental data to generate metadata, time stamping the metadata using the capture time data; receiving video data and video time data at a processor; correlating the metadata to the video data using the capture time data and the video time data; receiving a search query; and/or identifying a frame within the video data by performing a search of the metadata using the search criterion.

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BRIEF DESCRIPTION OF THE SEVERAL
VIEWS OF THE DRAWINGS

To easily identify the discussion of any particular element or act, the most significant digit or digits in a reference number refer to the figure number in which that element is first introduced.

FIG. 1 is a block diagram showing an example messaging system for exchanging data (e.g., messages and associated content) over a network.

FIG. 2 is block diagram illustrating further details regarding a messaging system, according to example embodiments.

FIG. 3 is a schematic diagram illustrating data which may be stored in the database of the messaging server system, according to certain example embodiments.

FIG. 4 is a schematic diagram illustrating a structure of a message, according to some embodiments, generated by a messaging client application for communication.

FIG. 5 is a schematic diagram illustrating an example access-limiting process, in terms of which access to content (e.g., an ephemeral message, and associated multimedia payload of data) or a content collection (e.g., an ephemeral message story) may be time-limited (e.g., made ephemeral).

FIG. 6 is a schematic diagram illustrating various hardware components of a messaging system, according to some example embodiments.

FIG. 7 is a schematic diagram illustrating various processors that may be implemented as part of a search processor, according to certain example embodiments.

FIG. 8 is a schematic diagram showing further architectural details of audio processors, according to some examples.

FIG. 9 is a schematic diagram illustrating the association of time information with environmental information, according to some examples.

FIG. 10 is a flowchart illustrating a method, according to some examples, of processing image data.

FIG. 11 is a flowchart illustrating a method, according to some examples, of performing an image-based search.

FIG. 12 is a schematic diagram showing an example environment within which a method to perform location-based content retrieval may be practiced and deployed.

FIG. 13 is a flowchart illustrating a method, according to some examples, of performing a multimedia-content based search.

FIG. 14 is a user interface diagram showing a search refinement user interface, according to some examples.

FIG. 15 is a user interface diagram showing a search interface, presented on a client device, according to some examples.

FIG. 16 is a block diagram illustrating a representative software architecture, which may be used in conjunction with various hardware architectures herein described.

FIG. 17 is a block diagram illustrating components of a machine, according to some example embodiments, able to read instructions from a machine-readable medium (e.g., a machine-readable storage medium) and perform any one or more of the methodologies discussed herein.

DETAILED DESCRIPTION

Glossary

“CARRIER SIGNAL” in this context refers to any intangible medium that is capable of storing, encoding, or carrying instructions for execution by the machine, and includes digital or analog communications signals or other

intangible medium to facilitate communication of such instructions. Instructions may be transmitted or received over the network using a transmission medium via a network interface device and using any one of a number of well-known transfer protocols.

“CLIENT DEVICE” in this context refers to any machine that interfaces to a communications network to obtain resources from one or more server systems or other client devices. A client device may be, but is not limited to, a mobile phone, desktop computer, laptop, portable digital assistants (PDAs), smart phones, tablets, ultra books, netbooks, laptops, multi-processor systems, microprocessor-based or programmable consumer electronics, game consoles, set-top boxes, or any other communication device that a user may use to access a network.

“COMMUNICATIONS NETWORK” in this context refers to one or more portions of a network that may be an ad hoc network, an intranet, an extranet, a virtual private network (VPN), a local area network (LAN), a wireless LAN (WLAN), a wide area network (WAN), a wireless WAN (WWAN), a metropolitan area network (MAN), the Internet, a portion of the Internet, a portion of the Public Switched Telephone Network (PSTN), a plain old telephone service (POTS) network, a cellular telephone network, a wireless network, a Wi-Fi® network, another type of network, or a combination of two or more such networks. For example, a network or a portion of a network may include a wireless or cellular network and the coupling may be a Code Division Multiple Access (CDMA) connection, a Global System for Mobile communications (GSM) connection, or other type of cellular or wireless coupling. In this example, the coupling may implement any of a variety of types of data transfer technology, such as Single Carrier Radio Transmission Technology (1xRTT), Evolution-Data Optimized (EVDO) technology, General Packet Radio Service (GPRS) technology, Enhanced Data rates for GSM Evolution (EDGE) technology, third Generation Partnership Project (3GPP) including 3G, fourth generation wireless (4G) networks, Universal Mobile Telecommunications System (UMTS), High Speed Packet Access (HSPA), Worldwide Interoperability for Microwave Access (WiMAX), Long Term Evolution (LTE) standard, others defined by various standard setting organizations, other long range protocols, or other data transfer technology.

“EPHEMERAL MESSAGE” in this context refers to a message that is accessible for a time-limited duration. An ephemeral message may be a text, an image, a video and the like. The access time for the ephemeral message may be set by the message sender. Alternatively, the access time may be a default setting or a setting specified by the recipient. Regardless of the setting technique, the message is transitory.

“MACHINE-READABLE MEDIUM” in this context refers to a component, device or other tangible media able to store instructions and data temporarily or permanently and may include, but is not be limited to, random-access memory (RAM), read-only memory (ROM), buffer memory, flash memory, optical media, magnetic media, cache memory, other types of storage (e.g., Erasable Programmable Read-Only Memory (EEPROM)) and/or any suitable combination thereof. The term “machine-readable medium” should be taken to include a single medium or multiple media (e.g., a centralized or distributed database, or associated caches and servers) able to store instructions. The term “machine-readable medium” shall also be taken to

by a machine, such that the instructions, when executed by one or more processors of the machine, cause the machine to perform any one or more of the methodologies described herein. Accordingly, a “machine-readable medium” refers to a single storage apparatus or device, as well as “cloud-based” storage systems or storage networks that include multiple storage apparatus or devices. The term “machine-readable medium” excludes signals per se.

“MODULE” in this context refers to logic having boundaries defined by function or subroutine calls, branch points, application program interfaces (APIs), or other technologies that provide for the partitioning or modularization of particular processing or control functions. Modules are typically combined via their interfaces with other modules to carry out a machine process. A module may be a packaged functional hardware unit designed for use with other components and a part of a program that usually performs a particular function of related functions. Modules may constitute either software modules (e.g., code embodied on a machine-readable medium) or hardware modules. A “hardware module” is a tangible unit capable of performing certain operations and may be configured or arranged in a certain physical manner. In various example embodiments, one or more computer systems (e.g., a standalone computer system, a client computer system, or a server computer system) or one or more hardware modules of a computer system (e.g., a processor or a group of processors) may be configured by software (e.g., an application or application portion) as a hardware module that operates to perform certain operations as described herein. In some embodiments, a hardware module may be implemented mechanically, electronically, or any suitable combination thereof. For example, a hardware module may include dedicated circuitry or logic that is permanently configured to perform certain operations. For example, a hardware module may be a special-purpose processor, such as a Field-Programmable Gate Array (FPGA) or an Application Specific Integrated Circuit (ASIC). A hardware module may also include programmable logic or circuitry that is temporarily configured by software to perform certain operations. For example, a hardware module may include software executed by a general-purpose processor or other programmable processor. Once configured by such software, hardware modules become specific machines (or specific components of a machine) uniquely tailored to perform the configured functions and are no longer general-purpose processors. It will be appreciated that the decision to implement a hardware module mechanically, in dedicated and permanently configured circuitry, or in temporarily configured circuitry (e.g., configured by software) may be driven by cost and time considerations. Accordingly, the phrase “hardware module” (or “hardware-implemented module”) should be understood to encompass a tangible entity, be that an entity that is physically constructed, permanently configured (e.g., hardwired), or temporarily configured (e.g., programmed) to operate in a certain manner or to perform certain operations described herein. Considering embodiments in which hardware modules are temporarily configured (e.g., programmed), each of the hardware modules need not be configured or instantiated at any one instance in time. For example, where a hardware module comprises a general-purpose processor configured by software to become a special-purpose processor, the general-purpose processor may be configured as respectively different special-purpose processors (e.g., comprising different hardware modules) at different times. Software accordingly configures a particular processor or processors, for example, to constitute a par-

ticular hardware module at one instance of time and to constitute a different hardware module at a different instance of time. Hardware modules can provide information to, and receive information from, other hardware modules. Accordingly, the described hardware modules may be regarded as being communicatively coupled. Where multiple hardware modules exist contemporaneously, communications may be achieved through signal transmission (e.g., over appropriate circuits and buses) between or among two or more of the hardware modules. In embodiments in which multiple hardware modules are configured or instantiated at different times, communications between such hardware modules may be achieved, for example, through the storage and retrieval of information in memory structures to which the multiple hardware modules have access. For example, one hardware module may perform an operation and store the output of that operation in a memory device to which it is communicatively coupled. A further hardware module may then, at a later time, access the memory device to retrieve and process the stored output. Hardware modules may also initiate communications with input or output devices, and can operate on a resource (e.g., a collection of information). The various operations of example methods described herein may be performed, at least partially, by one or more processors that are temporarily configured (e.g., by software) or permanently configured to perform the relevant operations. Whether temporarily or permanently configured, such processors may constitute processor-implemented modules that operate to perform one or more operations or functions described herein. As used herein, “processor-implemented module” refers to a hardware module implemented using one or more processors. Similarly, the methods described herein may be at least partially processor-implemented, with a particular processor or processors being an example of hardware. For example, at least some of the operations of a method may be performed by one or more processors or processor-implemented modules. Moreover, the one or more processors may also operate to support performance of the relevant operations in a “cloud computing” environment or as a “software as a service” (SaaS). For example, at least some of the operations may be performed by a group of computers (as examples of machines including processors), with these operations being accessible via a network (e.g., the Internet) and via one or more appropriate interfaces (e.g., an Application Program Interface (API)). The performance of certain of the operations may be distributed among the processors, not only residing within a single machine, but deployed across a number of machines. In some example embodiments, the processors or processor-implemented modules may be located in a single geographic location (e.g., within a home environment, an office environment, or a server farm). In other example embodiments, the processors or processor-implemented modules may be distributed across a number of geographic locations.

“PROCESSOR” in this context refers to any circuit or virtual circuit (a physical circuit emulated by logic executing on an actual processor) that manipulates data values according to control signals (e.g., “commands”, “op codes”, “machine code”, etc.) and which produces corresponding output signals that are applied to operate a machine. A processor may, for example, be a Central Processing Unit (CPU), a Reduced Instruction Set Computing (RISC) processor, a Complex Instruction Set Computing (CISC) processor, a Graphics Processing Unit (GPU), a Digital Signal Processor (DSP), an Application Specific Integrated Circuit (ASIC), a Radio-Frequency Integrated Circuit (RFIC) or any combination thereof. A processor may further be a multi-

core processor having two or more independent processors sometimes referred to as “cores”) that may execute instructions contemporaneously.

“TIMESTAMP” in this context refers to a sequence of characters or encoded information identifying when a certain event occurred, for example giving date and time of day, sometimes accurate to a small fraction of a second.

Description

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The description that follows includes systems, methods, techniques, instruction sequences, and computing machine program products that embody illustrative embodiments of the disclosure. In the following description, for the purposes of explanation, numerous specific details are set forth in order to provide an understanding of various embodiments of the inventive subject matter. It will be evident, however, to those skilled in the art, that embodiments of the inventive subject matter may be practiced without these specific details. In general, well-known instruction instances, protocols, structures, and techniques are not necessarily shown in detail.

Drawings

FIG. 1 is a block diagram showing an example messaging system **100** for exchanging data (e.g., messages and associated content) over a network. The messaging system **100** includes multiple client devices **102**, each of which hosts a number of applications including a messaging client application **104**. Each messaging client application **104** is communicatively coupled to other instances of the messaging client application **104** and a messaging server system **108** via a network **106** (e.g., the Internet).

Accordingly, each messaging client application **104** is able to communicate and exchange data with another messaging client application **104** and with the messaging server system **108** via the network **106**. The data exchanged between messaging client applications **104**, and between a messaging client application **104** and the messaging server system **108**, includes functions (e.g., commands to invoke functions) as well as payload data (e.g., text, audio, video or other multimedia data).

The messaging server system **108** provides server-side functionality via the network **106** to a particular messaging client application **104**. While certain functions of the messaging system **100** are described herein as being performed by either a messaging client application **104** or by the messaging server system **108**, it will be appreciated that the location of certain functionality either within the messaging client application **104** or the messaging server system **108** is a design choice. For example, it may be technically preferable to initially deploy certain technology and functionality within the messaging server system **108**, but to later migrate this technology and functionality to the messaging client application **104** where a client device **102** has a sufficient processing capacity.

The messaging server system **108** supports various services and operations that are provided to the messaging client application **104**. Such operations include transmitting data to, receiving data from, and processing data generated

by the messaging client application **104**. This data may include, message content, client device information, geolocation information, media annotation and overlays, message content persistence conditions, social network information, and live event information, as examples. Data exchanges within the messaging system **100** are invoked and controlled through functions available via user interfaces (UIs) of the messaging client application **104**.

Turning now specifically to the messaging server system **108**, an Application Program Interface (API) server **110** is coupled to, and provides a programmatic interface to, an application server **112**. The application server **112** is communicatively coupled to a database server **118**, which facilitates access to a database **120** in which is stored data associated with messages processed by the application server **112**.

Dealing specifically with the Application Program Interface (API) server **110**, this server receives and transmits message data (e.g., commands and message payloads) between the client device **102** and the application server **112**. Specifically, the Application Program Interface (API) server **110** provides a set of interfaces (e.g., routines and protocols) that can be called or queried by the messaging client application **104** in order to invoke functionality of the application server **112**. The Application Program interface (API) server **110** exposes various functions supported by the application server **112**, including account registration, login functionality, the sending of messages, via the application server **112**, from a particular messaging client application **104** to another messaging client application **104**, the sending of media files (e.g., images or video) from a messaging client application **104** to the messaging server application **114**, and for possible access by another messaging client application **104**, the setting of a collection of media data (e.g., story), the retrieval of a list of friends of a user of a client device **102**, the retrieval of such collections, the retrieval of messages and content, the adding and deletion of friends to a social graph, the location of friends within a social graph, opening and application event (e.g., relating to the messaging client application **104**).

The application server **112** hosts a number of applications and subsystems, including a messaging server application **114**, an image processing system **116** and a social network system **122**. The messaging server application **114** implements a number of message processing technologies and functions, particularly related to the aggregation and other processing of content (e.g., textual and multimedia content) included in messages received from multiple instances of the messaging client application **104**. As will be described in further detail, the text and media content from multiple sources may be aggregated into collections of content (e.g., called stories or galleries). These collections are then made available, by the messaging server application **114**, to the messaging client application **104**. Other processor and memory intensive processing of data may also be performed server-side by the messaging server application **114**, in view of the hardware requirements for such processing.

The application server **112** also includes an image processing system **116** that is dedicated to performing various image processing operations, typically with respect to images or video received within the payload of a message at the messaging server application **114**.

The social network system **122** supports various social networking functions services, and makes these functions and services available to the messaging server application **114**. To this end, the social network system **122** maintains and accesses an entity graph **304** within the database **120**.

Examples of functions and services supported by the social network system **122** include the identification of other users of the messaging system **100** with which a particular user has relationships or is “following”, and also the identification of other entities and interests of a particular user.

The application server **112** is communicatively coupled to a database server **118**, which facilitates access to a database **120** in which is stored data associated with messages processed by the messaging server application **114**.

FIG. 2 is block diagram illustrating further details regarding the messaging system **100**, according to example embodiments. Specifically, the messaging system **100** is shown to comprise the messaging client application **104** and the application server **112**, which in turn embody a number of some subsystems, namely an ephemeral timer system **202**, a collection management system **204** and an annotation system **206**.

The ephemeral timer system **202** is responsible for enforcing the temporary access to content permitted by the messaging client application **104** and the messaging server application **114**. To this end, the ephemeral timer system **202** incorporates a number of timers that, based on duration and display parameters associated with a message, or collection of messages (e.g., a SNAPCHAT story), selectively display and enable access to messages and associated content via the messaging client application **104**. Further details regarding the operation of the ephemeral timer system **202** are provided below.

The collection management system **204** is responsible for managing collections of media (e.g., collections of text, image video and audio data). In some examples, a collection of content (e.g., messages, including images, video, text and audio) may be organized into an “event gallery” or an “event story.” Such a collection may be made available for a specified time period, such as the duration of an event to which the content relates. For example, content relating to a music concert may be made available as a “story” for the duration of that music concert. The collection management system **204** may also be responsible for publishing an icon that provides notification of the existence of a particular collection to the user interface of the messaging client application **104**.

The collection management system **204** furthermore includes a curation interface **208** that allows a collection manager to manage and curate a particular collection of content. For example, the curation interface **208** enables an event organizer to curate a collection of content relating to a specific event (e.g., delete inappropriate content or redundant messages). Additionally, the collection management system **204** employs machine vision (or image recognition technology) and content rules to automatically curate a content collection. In certain embodiments, compensation may be paid to a user for inclusion of user generated content into a collection. In such cases, the curation interface **208** operates to automatically make payments to such users for the use of their content.

The annotation system **206** provides various functions that enable a user to annotate or otherwise modify or edit media content associated with a message. For example, the annotation system **206** provides functions related to the generation and publishing of media overlays for messages processed by the messaging system **100**. The annotation system **206** operatively supplies a media overlay (e.g., a SNAPCHAT filter) to the messaging client application **104** based on a geolocation of the client device **102**. In another example, the annotation system **206** operatively supplies a media overlay to the messaging client application **104** based

on other information, such as, social network information of the user of the client device **102**. A media overlay may include audio and visual content and visual effects. Examples of audio and visual content include pictures, texts, logos, animations, and sound effects. An example of a visual effect includes color overlaying. The audio and visual content or the visual effects can be applied to a media content item (e.g., a photo) at the client device **102**. For example, the media overlay including text that can be overlaid on top of a photograph generated taken by the client device **102**. In another example, the media overlay includes an identification of a location overlay (e.g., Venice beach), a name of a live event, or a name of a merchant overlay (e.g., Beach Coffee House). In another example, the annotation system **206** uses the geolocation of the client device **102** to identify a media overlay that includes the name of a merchant at the geolocation of the client device **102**. The media overlay may include other indicia associated with the merchant. The media overlays may be stored in the database **120** and accessed through the database server **118**.

In one example embodiment, the annotation system **206** provides a user-based publication platform that enables users to select a geolocation on a map, and upload content associated with the selected geolocation. The user may also specify circumstances under which a particular media overlay should be offered to other users. The annotation system **206** generates a media overlay that includes the uploaded content and associates the uploaded content with the selected geolocation.

In another example embodiment, the annotation system **206** provides a merchant-based publication platform that enables merchants to select a particular media overlay associated with a geolocation via a bidding process. For example, the annotation system **206** associates the media overlay of a highest bidding merchant with a corresponding geolocation for a predefined amount of time.

FIG. 3 is a schematic diagram illustrating data **300**, which may be stored in the database **120** of the messaging server system **108**, according to certain example embodiments. While the content of the database **120** is shown to comprise a number of tables, it will be appreciated that the data could be stored in other types of data structures (e.g., as an object-oriented database).

The database **120** includes message data stored within a message table **314**. The entity table **302** stores entity data, including an entity graph **304**. Entities for which records are maintained within the entity table **302** may include individuals, corporate entities, organizations, objects, places, events etc. Regardless of type, any entity regarding which the messaging server system **108** stores data may be a recognized entity. Each entity is provided with a unique identifier, as well as an entity type identifier (not shown).

The entity graph **304** furthermore stores information regarding relationships and associations between entities. Such relationships may be social, professional (e.g., work at a common corporation or organization) interested-based or activity-based, merely for example.

The database **120** also stores annotation data, in the example form of filters, in an annotation table **312**. Filters for which data is stored within the annotation table **312** are associated with and applied to videos (for which data is stored in a video table **310**) and/or images (for which data is stored in an image table **308**). Filters, in one example, are overlays that are displayed as overlaid on an image or video during presentation to a recipient user. Filters may be of various types, including a user-selected filters from a gallery of filters presented to a sending user by the messaging client

application **104** when the sending user is composing a message. Other types of filters include geolocation filters (also known as geo-filters) which may be presented to a sending user based on geographic location. For example, geolocation filters specific to a neighborhood or special location may be presented within a user interface by the messaging client application **104**, based on geolocation information determined by a GPS unit of the client device **102**. Another type of filter is a data filter, which may be selectively presented to a sending user by the messaging client application **104**, based on other inputs or information gathered by the client device **102** during the message creation process. Example of data filters include current temperature at a specific location, a current speed at which a sending user is traveling, battery life for a client device **102** or the current time.

Other annotation data that may be stored within the image table **308** is so-called “lens” data. A “lens” may be a real-time special effect and sound that may be added to an image or a video.

As mentioned above, the video table **310** stores video data which, in one embodiment, is associated with messages for which records are maintained within the message table **314**. Similarly, the image table **308** stores image data associated with messages for which message data is stored in the entity table **302**. The entity table **302** may associate various annotations from the annotation table **312** with various images and videos stored in the image table **308** and the video table **310**.

A story table **306** stores data regarding collections of messages and associated image, video or audio data, which are compiled into a collection (e.g., a SNAPCHAT story or a gallery). The creation of a particular collection may be initiated by a particular user (e.g., each user for which a record is maintained in the entity table **302**). A user may create a “personal story” in the form of a collection of content that has been created and sent/broadcast by that user. To this end, the user interface of the messaging client application **104** may include an icon that is user selectable to enable a sending user to add specific content to his or her personal story.

A collection may also constitute a “live story,” which is a collection of content from multiple users that is created manually, automatically or using a combination of manual and automatic techniques. For example, a “live story” may constitute a curated stream of user-submitted content from various locations and events. Users, whose client devices have location services enabled and are at a common location event at a particular time may, for example, be presented with an option, via a user interface of the messaging client application **104**, to contribute content to a particular live story. The live story may be identified to the user by the messaging client application **104**, based on his or her location. The end result is a “live story” told from a community perspective.

A further type of content collection is known as a “location story”, which enables a user whose client device **102** is located within a specific geographic location (e.g., on a college or university campus) to contribute to a particular collection. In some embodiments, a contribution to a location story may require a second degree of authentication to verify that the end user belongs to a specific organization or other entity (e.g., is a student on the university campus).

FIG. 4 is a schematic diagram illustrating a structure of a message **400**, according to some in some embodiments, generated by a messaging client application **104** for communication to a further messaging client application **104** or

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the messaging server application 114. The content of a particular message 400 is used to populate the message table 314 stored within the database 120, accessible by the messaging server application 114. Similarly, the content of a message 400 is stored in memory as “in-transit” or “in-flight” data of the client device 102 or the application server 112. The message 400 is shown to include the following components:

A message identifier 402: a unique identifier that identifies the message 400.

A message text payload 404: text, to be generated by a user via a user interface of the client device 102 and that is included in the message 400.

A message image payload 406: image data, captured by a camera component of a client device 102 or retrieved from memory of a client device 102, and that is included in the message 400.

A message video payload 408: video data, captured by a camera component or retrieved from a memory component of the client device 102 and that is included in the message 400.

A message audio payload 410: audio data, captured by a microphone or retrieved from the memory component of the client device 102, and that is included in the message 400.

A message annotations 412: annotation data (e.g., filters, stickers or other enhancements) that represents annotations to be applied to message image payload 406, message video payload 408, or message audio payload 410 of the message 400.

A message duration parameter 414: parameter value indicating, in seconds, the amount of time for which content of the message (e.g., the message image payload 406, message video payload 408, message audio payload 410) is to be presented or made accessible to a user via the messaging client application 104.

A message geolocation parameter 416: geolocation data (e.g., latitudinal and longitudinal coordinates) associated with the content payload of the message. Multiple message geolocation parameter 416 values may be included in the payload, each of these parameter values being associated with respect to content items included in the content (e.g., a specific image into within the message image payload 406, or a specific video in the message video payload 408).

A message story identifier 418: identifier values identifying one or more content collections (e.g., “stories”) with which a particular content item in the message image payload 406 of the message 400 is associated. For example, multiple images within the message image payload 406 may each be associated with multiple content collections using identifier values.

A message tag 420: each message 400 may be tagged with multiple tags, each of which is indicative of the subject matter of content included in the message payload. For example, where a particular image included in the message image payload 406 depicts an animal (e.g., a lion), a tag value may be included within the message tag 420 that is indicative of the relevant animal. Tag values may be generated manually, based on user input, or may be automatically generated using, for example, image recognition.

A message sender identifier 422: an identifier a messaging system identifier, email address or device identifier) indicative of a user of the client device 102 on which the message 400 was generated and from which the message 400 was sent

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A message receiver identifier 424: an identifier (e.g., a messaging system identifier, email address or device identifier) indicative of a user of the client device 102 to which the message 400 is addressed.

The contents (e.g. values) of the various components of message 400 may be pointers to locations in tables within which content data values are stored. For example, an image value in the message image payload 406 may be a pointer to (or address of) a location within an image table 308. Similarly, values within the message video payload 408 may point to data stored within a video table 310, values stored within the message annotations 412 may point to data stored in an annotation table 312, values stored within the message story identifier 418 may point to data stored in a story table 306, and values stored within the message sender identifier 422 and the message receiver identifier 424 may point to user records stored within an entity table 302.

FIG. 5 is a schematic diagram illustrating an access-limiting process 500, in terms of which access to content (e.g., an ephemeral message 502, and associated multimedia payload of data) or a content collection (e.g., an ephemeral message story 504) may be time-limited (e.g., made ephemeral).

An ephemeral message 502 is shown to be associated with a message duration parameter 506, the value of which determines an amount of time that the ephemeral message 502 will be displayed to a receiving user of the ephemeral message 502 by the messaging client application 104. In one embodiment, where the messaging client application 104 is a SNAPCHAT application client, an ephemeral message 502 is viewable by a receiving user for up to a maximum of 10 seconds, depending on the amount of time that the sending user specifies using the message duration parameter 506.

The message duration parameter 506 and the message receiver identifier 424 are shown to be inputs to a message timer 512, which is responsible for determining the amount of time that the ephemeral message 502 is shown to a particular receiving user identified by the message receiver identifier 424. In particular, the ephemeral message 502 will only be shown to the relevant receiving user for a time period determined by the value of the message duration parameter 506. The message timer 512 is shown to provide output to a more generalized ephemeral timer system 202, which is responsible for the overall timing of display of content (e.g., an ephemeral message 502) to a receiving user.

The ephemeral message 502 is shown in FIG. 5 to be included within an ephemeral message story 504 (e.g., a personal SNAPCHAT story, or an event story). The ephemeral message story 504 has an associated story duration parameter 508, a value of which determines a time-duration for which the ephemeral message story 504 is presented and accessible to users of the messaging system 100. The story duration parameter 508, for example, may be the duration of a music concert, where the ephemeral message story 504 is a collection of content pertaining to that concert. Alternatively, a user (either the owning user or a curator user) may specify the value for the story duration parameter 508 when performing the setup and creation of the ephemeral message story 504.

Additionally, each ephemeral message 502 within the ephemeral message story 504 has an associated story participation parameter 510, a value of which determines the duration of time for which the ephemeral message 502 will be accessible within the context of the ephemeral message story 504. Accordingly, a particular ephemeral message story 504 may “expire” and become inaccessible within the context of the ephemeral message story 504, prior to the

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ephemeral message story 504 itself expiring in terms of the story duration parameter 508. The story duration parameter 508, story participation parameter 510, and message receiver identifier 424 each provide input to a story timer 514, which operationally determines, firstly, whether a particular ephemeral message 502 of the ephemeral message story 504 will be displayed to a particular receiving user and, if so, for how long. Note that the ephemeral message story 504 is also aware of the identity of the particular receiving user as a result of the message receiver identifier 424.

Accordingly, the story timer 514 operationally controls the overall lifespan of an associated ephemeral message story 504, as well as an individual ephemeral message 502 included in the ephemeral message story 504. In one embodiment, each and every ephemeral message 502 within the ephemeral message story 504 remains viewable and accessible for a time-period specified by the story duration parameter 508. In a further embodiment, a certain ephemeral message 502 may expire, within the context of ephemeral message story 504, based on a story participation parameter 510. Note that a message duration parameter 506 may still determine the duration of time for which a particular ephemeral message 502 is displayed to a receiving user, even within the context of the ephemeral message story 504. Accordingly, the message duration parameter 506 determines the duration of time that a particular ephemeral message 502 is displayed to a receiving user, regardless of whether the receiving user is viewing that ephemeral message 502 inside or outside the context of an ephemeral message story 504.

The ephemeral timer system 202 may furthermore operationally remove a particular ephemeral message 502 from the ephemeral message story 504 based on a determination that it has exceeded an associated story participation parameter 510. For example, when a sending user has established a story participation parameter 510 of 24 hours from posting, the ephemeral timer system 202 will remove the relevant ephemeral message 502 from the ephemeral message story 504 after the specified 24 hours. The ephemeral timer system 202 also operates to remove an ephemeral message story 504 either when the story participation parameter 510 for each and every ephemeral message 502 within the ephemeral message story 504 has expired, or when the ephemeral message story 504 itself has expired in terms of the story duration parameter 508.

In certain use cases, a creator of a particular ephemeral message story 504 may specify an indefinite story duration parameter 508. In this case, the expiration of the story participation parameter 510 for the last remaining ephemeral message 502 within the ephemeral message story 504 will determine when the ephemeral message story 504 itself expires. In this case, a new ephemeral message 502, added to the ephemeral message story 504, with a new story participation parameter 510, effectively extends the life of an ephemeral message story 504 to equal the value of the story participation parameter 510.

Responsive to the ephemeral timer system 202 determining that an ephemeral message story 504 has expired (e.g., is no longer accessible), the ephemeral timer system 202 communicates with the messaging system 100 (and, for example, specifically the messaging client application 104) to cause an indicium (e.g., an icon) associated with the relevant ephemeral message story 504 to no longer be displayed within a user interface of the messaging client application 104. Similarly, when the ephemeral timer system 202 determines that the message duration parameter 506 for a particular ephemeral message 502 has expired, the ephemeral

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timer system 202 causes the messaging client application 104 to no longer display an indicium (e.g., an icon or textual identification) associated with the ephemeral message 502.

FIG. 6 is a schematic diagram illustrating various hardware components of a messaging system 100, according to some example embodiments. As noted elsewhere herein, the messaging system 100 may include both the messaging client application 104 and the messaging server system 108. The location of certain of the components described with reference to FIG. 6 is a design choice, and fungible. Certain embodiments may be implemented as a peer-to-peer system, in which the messaging server system 108 provides relatively little processing. In these embodiments, the majority of the components described with reference to FIG. 6 may form part of a client device 102. Certain other embodiments may be implemented using a “thin client” architecture, in which case most of the processing power (and associated components) are located within an application server 112.

Referring specifically to FIG. 6, the messaging system 100 is shown to include various I/O components 1718 including a microphone 602, a camera 604, a GPS unit 606, an accelerometer 608 and an input device 624 (e.g., an alphanumeric keyboard or tactile touchscreen). Each of these components is coupled to a clock 610, which provides time information to these components. Accordingly, the various components are able to time stamp (or otherwise associate time information with) outputted data. For example, the clock 610 may output time and data information, or otherwise simple internal timing information, with which the various components are able to time stamp respective outputs.

The components provide respective outputs to appropriate input/output (I/O) interfaces. Specifically, the camera 604 is communicatively coupled to a video I/O interface 614, while the other components are shown coupled to further I/O interfaces 612. The camera 604, via the video I/O interface 614, outputs time-stamped video data 616 to an input port of a search processor 620. Similarly, the other components, via the I/O interfaces 612, output other types of environmental data 618 to respective input ports of the search processor 620. Specifically, the accelerometer 608 outputs motion data, the GPS unit 606 outputs location data (e.g., latitudinal and longitudinal coordinates), and the microphone 602 outputs audio data. Other I/O devices output other forms of environmental data 618.

FIG. 7 is a schematic diagram illustrating various processors 700 that may be implemented as part of the search processor 620. These additional processors may, in some example embodiments, be instantiated temporarily within the context of the search processor 620 by instructions received from storage or memory associated with the search processor 620. In other embodiments, these additional processors may be hardwired or permanently configured as components of the search processor 620 (e.g., as ASIC components).

The additional processors include image processors 702, audio processor 704, motion processors 706 and location processors 708. These processors 700, as will be described in further detail herein, process the environmental data received at the search processor 620 from the various I/O components of the messaging system 100. The processors 700 may also share a certain functionality and components.

The processors 700 are also shown to be communicatively coupled to each other, and to I/O pins of the search processor 620. In one embodiment, these couplings are implemented via a bus 710. Where multiple processors 700 exist contemporaneously,

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aneously, communications may be achieved through signal transmission (e.g., over appropriate circuits and buses) between or among two or more of the processors 700 and/or pins via the bus 710. In embodiments in which multiple processors 700 are configured or instantiated at different times, communications between the processors 700 may be achieved, for example, through the storage and retrieval of information in memory structures to which the processors 700 have access.

FIG. 8 is a schematic diagram showing further architectural details of an audio processor 704, according to some examples.

Specifically, the audio processor 704 includes a number of filters and sub-systems, namely a low-pass filter 802, a band-pass filter 804, a high-pass filter 806, an audio fingerprint component 808 and a voice recognition component 818. The audio processor 704 receives environmental data 618 in the form of audio data (e.g., from the microphone 602) collected from an environment (or context) in which the client device 102 is located. The filters are deployed to isolate various components of the audio data for further processing. The low-pass filter 802 operationally isolates background audio data, while the high-pass filter 806 is tuned to operationally isolate voice frequencies with the audio data. The band-pass filter 804 is tuned to isolate frequencies of audio data (e.g., a combination of frequencies) that may be used to generate contextual markers for a specific context or environment.

The audio fingerprint component 808 receives and processes isolated audio data from the filters, and operationally attempts to match audio fingerprints 812, stored in an associated storage 810, with hashed versions of the isolated audio data received from the filters. For example, the storage 810 may store a collection of audio fingerprints 812 related to a known location in a museum. In this example, a particular sound (e.g., a background recording or combination of environments sounds) is typically present at the know location, and an audio fingerprints 812 is generated based on the particular sound. Low-pass audio data received from the low-pass filter 802 is hashed and compared to the audio fingerprints 812 by the audio fingerprint component 808. An audio print match is then used to generate metadata, based on the audio data and the associated time stamps. From example, the metadata 816 may be generated to identify the museum location at which the matching audio fingerprints 812 were generated, as well as the names (or other identifiers) of displays, content and items located within a specific proximity of the museum location. This metadata 816 is then associated with corresponding video data 616, using time stamp data.

The voice recognition component 818 operationally receives audio data from either the band-pass filter 804 or the high-pass filter 806, and generates textual data based in the received audio data. For example, the voice recognition component 818 may include (or remotely access) voice recognition technology developed by NUANANCE TECHNOLOGIES in order generate textual data, which is time stamped to generate the metadata 816 outputted by the audio processor 704. For example, a user of the client device 102 may provide a verbal description via the microphone 602 of what is being recorded (or other relevant commentary) while recording specific video data 616. This captured audio data is then automatically transcribed by the voice recognition component 818 to generate the metadata 816, which is the associated with the specific video data 616.

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In the examples manners described above, video data 616 is associated with metadata 816 generated using audio data captured using the microphone 602 of a client device 102.

FIG. 9 is a schematic diagram illustrating the association of time information with environmental information, according to some examples.

As mentioned with respect to an earlier figure, video data 616 and environmental data 618 may be time stamped with time information received from the clock 610 by various components. To this end, FIG. 9 shows (1) frames of video data 616, each time stamped with a respective time stamps 906 and (2) environmental data 618, in the example form of instances of sensor data 908, each time stamped with a respective time stamps 910. The video data 616 and the environmental data 618 are shown to be input to the search processor 620. The search processor 620 uses the environmental data 618 to generate and output metadata 912, instances of which are also each time stamped with a respective time stamps 904. This metadata 912 is stored within memory or storage 810 (e.g., database 120), and is accessible by the search processor 620 for use in responding to a search query 902, responsive to which a search result 628 (e.g., including video data 616) is outputted.

The metadata 912 is associated with the video data 616, using respective time stamps 906 and time stamps 904, so that particular metadata 912 is associated with a particular frame or series of frames) of the video data 616. In one example, the metadata 912 may be incorporated into the message tag 420 of a message 400, so as to render the message 400 locatable by a search. In other embodiments, the metadata 912 may simply be associated with or related to the video data 616 by way of pointers.

In examples, the metadata 912 renders the video data 616 more richly searchable by providing text information that can be conveniently and easily searched in order to identify particular frames of the video data 616. The environmental data 618 may be used to generate a deep and rich body of metadata 912 that is temporally associated with the video data 616. To this end, the metadata 912 may provide a rich body of environmental and contextual information that is associated with the video data 616 in order to render it searchable in ways that has not previously been possible.

For example, using information received from a variety of sensors, a variety of different types of metadata 912 may be generated and associated with a particular frame or series of frames of the video data 616. Consider that information outputted from the accelerometer 608 may be used to generate metadata 912 providing contextual speed information, identifying the speed at which a mobile client device 102 was travelling when used to capture a particular set of frames of the video data 616.

Similarly, audio data may be processed to identify a wide variety of environmental and contextual characteristics applicable to the capture environment and context of a frame or series of frames of the video data 616. For example, the identification of background noise from airplanes may indicate that particular video data 616 was taken in geographic proximity to an airport. Background noise from trains may indicate that video data 616 was captured in a train station. Audio data may similarly be analyzed, using voice recognition technology, to identify a speaker, or specific content that was being broadcast (e.g., a broadcast television show) at a particular time. Music audio may also be analyzed to identify a particular song or band playing in the background while certain video data 616 was captured. The identification of such a song or band may allow the location and time at

which particular video data **616** were captured to be ascertained with a relatively high degree of accuracy and certainty.

Multiple types of environmental data **618** may also be combined to increase certainty regarding the validity and applicability of some assumptions made during the generation of metadata **912**. For example, a high degree of correlation between location data, outputted by a GPS unit **606**, and audio data outputted by a microphone **602**, allows a high degree of certainty to be attributed to certain metadata **912** (e.g., that a video was captured at a specific music concert on a specific date).

FIG. **10** is a flowchart illustrating a method **1000**, according to some examples, of processing image data.

The method **1000** commences at block **1002** with the receiving of environmental data and associated capture time data from the sensors of a mobile computing device. With specific reference to FIG. **6**, environmental data **618** may be received at the search processor **620** from any one of the I/O devices of the messaging system **100**, including speed information from the accelerometer **608**, location information from the GPS unit **606**, and audio data from the microphone **602**. The environmental data **618** is received at the search processor **620**, via the I/O interfaces **612**. In some examples, with reference to FIG. **9**, the environmental data is sensor data **908**, instances of which are time stamped with a respective time stamps **910**.

At block **1004**, the search processor **620** operates to generate metadata **912**, using the environmental data **618** as input. In one example, the environmental data **618** is audio data, and the search processor **620** processes this environmental data **618** in order to identify a sound within the audio data that is associated with a place, object, event or entity. To this end, the audio processor **704** of the search processor **620** are deployed to isolate and extract specific frequencies or sound types from the audio data. The low-pass filter **802**, the band-pass filter **804**, and the high-pass filter **806** are employed, as described above with reference to FIG. **8**, to filter the received audio data to generate filtered audio data. The low-pass filter **802** may be used to extract background audio data which is then provided as input to the audio fingerprint component **808** for further processing.

The audio processor **704** uses fingerprinting technology (e.g., the audio fingerprint component **808**, which operatively generates audio fingerprints from the received audio data and compares these fingerprints to the audio fingerprints **812** stored in storage **810**) to identify places, objects, events and entities.

The audio processor **704** also use the voice recognition component **818** to perform voice recognition and generate textual data (e.g., a collection of words) from the audio data.

At block **1006**, the search processor **620** time stamps the metadata **912** with time stamps **904** (e.g., corresponding to the time stamps **910**).

At block **1008**, the search processor **620** receives the video data **616**, from the camera **604** via the video I/O interface **614**, for processing. At block **1010**, the search processor **620** correlates the video data **616** with the metadata **912**, using the time stamps **906** and the time stamps **904**. In one embodiment, the video data **616** comprises video data included in a message **400** (e.g., as message video payload **408**), and the message tag **420** of the relevant message **400** is updated to include the metadata **912**. In a further embodiment, a pointer in the message tag **420** is updated to identify a particular instance of the metadata **912**, based on a correlation of the time stamps **906** and the time stamps **904**.

At block **1012**, the search processor **620** receives a search query **902**. The search query **902** includes search criteria, which may include a textual search criterion (e.g., a word or collection of keywords for text-based searching), an audio search criterion (e.g., for audio-based searching), or an image search criterion (e.g., an image or video data, for image-based searching).

The search processor **620** then performs the appropriate processing of the search query **902**, depending on the type and form of the search criteria. For example, where the search criteria includes an image, the image may be analyzed using the image processors **702** to perform entity identification (e.g., object recognition) on the image. Alternatively, an image included in the search criteria may be provided to a third party provider, which then returns keywords that form of the basis for a text-based search.

At block **1014**, the search processor **620** identifies a frame, or sequence of frames, within the video data **616** by performing a search of the metadata **912**, using the search criterion (or information derived from the search criteria). The search processor **620** may perform a text-based search of the metadata **912** to identify a correspondence between the search criterion and information included in the metadata **912**. Having identified one or more instances of the metadata **912** that show a predetermined degree of correlation or correspondence to the search criterion, the search processor **620** then identifies one or more frames of the video data **616** that correlate to the identified metadata **912**. In one specific example, the search processor **620** uses keywords, derived from the search criterion of the search query **902**, to search for a corresponding message tag **420** of a message **400** stored in the message table **314**. In this way, video data, stored in the video table **310** and identified by message video payload **408**, maybe identified at block **1014**.

The identified frames of the video data **616** are outputted as the search result **628** and displayed on a user interface of the client device **102**. The method **1000** then terminates at done block **1016**.

FIG. **11** is a flowchart illustrating a method **1100**, according to some examples, of performing an image-based search.

The method **1100** commences at block **1102**, with the receipt of a number of content items (e.g., images) from a content owner **1222** that is associated with a particular entity or location. FIG. **12** is a schematic diagram showing an example environment **1200** within which the method **1100** may be practiced and deployed.

Referring now to FIG. **12**, in one example, a particular location **1204** within a museum **1202** has multiple content items associated therewith by a content owner **1222** that together constitute an index **1210**. The location **1206** and location **1208** have similar indexes (not shown). Specifically, the location **1204** has the index **1210** including location data **1212**, audio data **1214**, image data **1216**, and text data **1218** associated therewith. In this example, the location **1204** may house an exhibit dealing with a specific period or event in the American Civil War. Here, the location data **1212** specifies the exact UPS coordinates of the exhibit. Instances of the audio data **1214** constitute recordings made at the exhibit, and include both background noise, as well as other ambient noise such as the soundtrack of a video that is part of the exhibit. Instances of the image data **1216** include various images or photographs taken of the exhibit, as well as the surroundings of the exhibit. In one example, the image data **1216** may include a 360° image taken using a specifically adapted camera at the location **1204**. The image data **1216** also includes videos captured at the location **1204**. Finally, instances of the text data **1218** include textual

descriptions of the exhibit, as well as words that are commonly and frequently spoken by either guides or visitors in proximity to the exhibit at the location **1204**. As such, the various content items relate to a common entity or event (e.g., the exhibit, or an event depicted or related to the exhibit).

Returning to FIG. **11**, at block **1104** the index **1210** of the various content items (e.g., location data **1212**, audio data **1214**, and image data **1216**) is created. In some example environments, this index **1210** is created by the search processor **620**, and are stored as metadata **912**.

At block **1106**, a further image is received from a mobile computing device of a user. In one embodiment, this further images is received as part of a search query **902** by the search processor **620**. In addition to the further image, the search query received at block **1106** may also include textual data (e.g., keywords) and audio data (e.g., audio data captured at a specific location of a client device **102**).

At block **1108**, the search processor **620** determines a correlation between this further image, received in the search query **902**, and content of the index **1210** (e.g., the metadata **912**). Others correlation may be detected by expanding the search query **902** based on the image included therein. For example, object recognition may be performed with respect to the further image to generate a set of keywords, which are then included in the search query **902**. Additionally, audio processing (as described above with reference to FIG. **8**) may be performed to also expand the search query based on audio data included therein.

At block **1110**, responsive to determining that a predetermined degree of correlation between the further image (and other content) and the index **1210**, the mobile computing device of the user may be directed to content of the content owner **1222**. Specifically, referring again to FIG. **12**, the index **1210** is mapped by maps **1224** to a collection of content items **1220**. The content items **1220** constitute further media and content relevant to the location **1204**. Again, where the location **1204** houses a museum exhibit, the content items **1220** are various multimedia content items (e.g., a publication, web page, video data, audio data etc.) that relate to, and supplement, the exhibit. In this way, a visitor to the location **1204** may initiate a query or search using environmental and/or contextual information of the exhibit, in order to identify content items **1220** relevant to that exhibit in a convenient and automatic way.

The method **1100** then terminates at done block **1112**.

FIG. **13** is a flowchart illustrating a method **1300**, according to some examples, of performing a multimedia-content based search. The method **1300** will be described with reference to both the flowchart shown in FIG. **13**, and the user interface shown in FIG. **14**. FIG. **14** is a user interface diagram showing a search refinement user interface **1400**, according to some examples.

The method **1300** commences at block **1302** with the receipt of a query image from a mobile computing device of a user. The query image may be received as part of a search query **902**, received at the search processor **620** of the messaging system **100**. Referring to FIG. **14**, the query image may be received from the camera of the client device **102**, or may be selected from storage via a user interface presented by the client device **102**. The search refinement user interface **1400** of FIG. **14** shows a basketball player image **1402** depicting a basketball player dribbling a basketball. User selection of the basketball player image **1402** by a search initiation action **1404** initiates a search query

generation process, by which the query image (e.g., the basketball player image **1402**) is provided to the search processor **620**.

At block **1304**, the search processor **620** uses the query image to conduct a content search to identify further content having a predetermined degree of relatedness to the query image. In one example, the search processor **620** conducts a search of metadata **912** that is associated with further content (e.g., that forms part of a message **400**, and is stored in the image table **308**, video table **310**, or annotation table **312**). This search of further content may also include a search of third-party content, and accordingly extend beyond content accessible within the messaging system **100**.

Referring to FIG. **14**, the search processor **620** processes the basketball player image **1402** to identify different components or aspects thereof. To this end, object recognition may be performed by the image processors **702** to parse the basketball player image **1402** and identify different objects contained within the basketball player image **1402**. Processing of the basketball player image **1402** may identify a basketball and basketball clothing, including basketball sneakers. The search query **902** is expanded, based on a number of different objects identified within the basketball player image **1402**. Accordingly, the search query **902** may be expanded, responsive to selection of the basketball player image **1402**, to search for basketballs, basketball sneakers, and basketball clothing.

At block **1306**, the search processor **620** outputs a search result **628**, which is presented as search results of the content search to a user by a mobile computing device (e.g., the client device **102**). As shown in FIG. **14**, the presented search results may thus include a sneaker image **1406**, a clothing image **1408**, and a basketball image **1410**.

At block **1308**, selection of a specific search result, from the set of search results, is received via the mobile computing device. For example, an image selection action **1412** of the basketball image **1410** triggers receipt of the basketball image **1410** by the search processor **620**.

At block **1310**, further content associated with the specific result is accessed. For example, the search processor **620**, responsive to receipt of the basketball image **1410**, generates a refining further search query **902** at block **1312**, initiates a further search, using the refining search query **902**, at block **1314**. As shown in FIG. **14**, this further search, performed responsive to selection of the basketball image **1410**, generates a refined search result set including basketball images **1414**, **1416**, and **1418**, which are presented within the search refinement user interface **1400** to a user. The method **1300** then terminates at done block **1316**.

Referring to FIG. **14** in particular, it will be appreciated that the method **1300** enables a user to perform an iterative, refining image-based search.

FIG. **15** is a user interface diagram showing a search interface, presented on a client device **102**, according to some examples. The interface includes a reference search query image **1502**. The query image **1502** may be captured via the camera **604** of the client device **102** or selected from images stored in local storage or remote storage accessible by the messaging client application **104** executing on the client device **102**.

The query image **1502** is processed by the image processors **702** of the search processor **620** to generate a collection of search terms **1504**. In one example, local image recognition is used to generate the search terms **1504**. In another example, a third-party application or service provided may be queried using the query image **1502** in order to receive the search terms **1504**. Certain (e.g., one, a sub-set or all) of

the search terms **1504** may be designated as “active” or “inactive”, either automatically by the messaging client application **104** or manually by a user.

The active search terms **1504** are then used to query a content repository to retrieve and present image results **1506** and video results **1508** within the context of the user interface. The image results **1506** and the video results **1508** may be one or more collections of content (e.g., SNAPCHAT Stories). By varying the active search terms **1504**, a user may cause different image results **1506** and video results **1508** to be presented and displayed.

FIG. **16** is a block diagram illustrating an example software architecture **1606**, which may be used in conjunction with various hardware architectures herein described. FIG. **16** is an example of a software architecture and it will be appreciated that many other architectures may be implemented to facilitate the functionality described herein. The software architecture **1606** may execute on hardware such as machine **1700** of FIG. **17** that includes, among other things, processors **1704**, memory **1714**, and I/O components **1718**. A representative hardware layer **1652** is illustrated and can represent, for example, the machine **1700** of FIG. **17**. The representative hardware layer **1652** includes a processing unit **1654** having associated executable instructions **1604**. Executable instructions **1604** represent the executable instructions of the software architecture **1606**, including implementation of the methods, modules and so forth described herein. The hardware layer **1652** also includes memory and/or storage modules memory/storage **1656**, which also have executable instructions **1604**. The hardware layer **1652** may also comprise other hardware **1658**.

In the example architecture of FIG. **16**, the software architecture **1606** may be conceptualized as a stack of layers where each layer provides particular functionality. For example, the software architecture **1606** may include layers such as an operating system **1602**, libraries **1620**, applications **1616** and a presentation layer **1614**. Operationally, the applications **1616** and/or other components within the layers may invoke application programming interface (API) API calls **1608** through the software stack and receive a response as in response to the API calls **1608**. The layers illustrated are representative in nature and not all software architectures have all layers. For example, some mobile or special purpose operating systems may not provide a frameworks/middleware **1618**, while others may provide such a layer. Other software architectures may include additional or different layers.

The operating system **1602** may manage hardware resources and provide common services. The operating system **1602** may include, for example, a kernel **1622**, services **1624** and drivers **1626**. The kernel **1622** may act as an abstraction layer between the hardware and the other software layers. For example, the kernel **1622** may be responsible for memory management, processor management (e.g., scheduling), component management, networking, security settings, and so on. The services **1624** may provide other common services for the other software layers. The drivers **1626** are responsible for controlling or interfacing with the underlying hardware. For instance, the drivers **1626** include display drivers, camera drivers, Bluetooth® drivers, flash memory drivers, serial communication drivers (e.g., Universal Serial Bus (USB) drivers), Wi-Fi® drivers, audio drivers, power management drivers, and so forth depending on the hardware configuration.

The libraries **1620** provide a common infrastructure that is used by the applications **1616** and/or other components and/or layers. The libraries **1620** provide functionality that

allows other software components to perform tasks in an easier fashion than to interface directly with the underlying operating system **1602** functionality (e.g., kernel **1622**, services **1624** and/or drivers **1626**). The libraries **1620** may include system libraries **1644** (e.g., C standard library) that may provide functions such as memory allocation functions, string manipulation functions, mathematical functions, and the like. In addition, the libraries **1620** may include API libraries **1646** such as media libraries (e.g., libraries to support presentation and manipulation of various media format such as MPREG4, H.264, MP3, AAC, AMR, JPG, PNG), graphics libraries (e.g., an OpenGL framework that may be used to render 2D and 3D in a graphic content on a display), database libraries (e.g., SQLite that may provide various relational database functions), web libraries (e.g., WebKit that may provide web browsing functionality), and the like. The libraries **1620** may also include a wide variety of other libraries **1648** to provide many other APIs to the applications **1616** and other software components/modules.

The frameworks/middleware **1618** (also sometimes referred to as middleware) provide a higher-level common infrastructure that may be used by the applications **1616** and/or other software components/modules. For example, the frameworks/middleware **1618** may provide various graphic user interface (GUI) functions, high-level resource management, high-level location services, and so forth. The frameworks/middleware **1618** may provide a broad spectrum of other APIs that may be utilized by the applications **1616** and/or other software components/modules, some of which may be specific to a particular operating system **1602** or platform.

The applications **1616** include built-in applications **1638** and/or third-party applications **1640**. Examples of representative built-in applications **1638** may include, but are not limited to, a contacts application, a browser application, a book reader application, a location application, a media application, a messaging application, and/or a game application. Third-party applications **1640** may include an application developed using the ANDROID™ or IOS™ software development kit (SDK) by an entity other than the vendor of the particular platform, and may be mobile software running on a mobile operating system such as IOS™, ANDROID™, WINDOWS® Phone, or other mobile operating systems. The third-party applications **1640** may invoke the API calls **1608** provided by the mobile operating system (such as operating system **1602**) to facilitate functionality described herein.

The applications **1616** may use built in operating system functions (e.g., kernel **1622**, services **1624** and/or drivers **1626**), libraries **1620**, and frameworks/middleware **1618** to create user interfaces to interact with users of the system. Alternatively, or additionally, in some systems interactions with a user may occur through a presentation layer, such as presentation layer **1614**. In these systems, the application/module “logic” can be separated from the aspects of the application/module that interact with a user.

FIG. **17** is a block diagram illustrating components of a machine **1700**, according to some example embodiments, able to read instructions from a machine-readable medium (e.g., a machine-readable storage medium) and perform any one or more of the methodologies discussed herein. Specifically, FIG. **17** shows a diagrammatic representation of the machine **1700** in the example form of a computer system, within which instructions **1710** (e.g., software, a program, an application, an applet, an app, or other executable code) for causing the machine **1700** to perform any one or more of the methodologies discussed herein may be

executed. As such, the instructions **1710** may be used to implement modules or components described herein. The instructions **1710** transform the general, non-programmed machine **1700** into a particular machine **1700** programmed to carry out the described and illustrated functions in the manner described. In alternative embodiments, the machine **1700** operates as a standalone device or may be coupled (e.g., networked) to other machines. In a networked deployment, the machine **1700** may operate in the capacity of a server machine or a client machine in a server-client network environment, or as a peer machine in a peer-to-peer (or distributed) network environment. The machine **1700** may comprise, but not be limited to, a server computer, a client computer, a personal computer (PC), a tablet computer, a laptop computer, a netbook, a set-top box (STB), a personal digital assistant (PDA), an entertainment media system, a cellular telephone, a smart phone, a mobile device, a wearable device (e.g., a smart watch), a smart home device (e.g., a smart appliance), other smart devices, a web appliance, a network router, a network switch, a network bridge, or any machine capable of executing the instructions **1710**, sequentially or otherwise, that specify actions to be taken by machine **1700**. Further, while only a single machine **1700** is illustrated, the term “machine” shall also be taken to include a collection of machines that individually or jointly execute the instructions **1710** to perform any one or more of the methodologies discussed herein.

The machine **1700** may include processors **1704**, memory memory/storage **1706**, and I/O components **1718**, which may be configured to communicate with each other such as via a bus **1702**. The memory/storage **1706** may include a memory **1714**, such as a main memory, or other memory storage, and a storage unit **1716**, both accessible to the processors **1704** such as via the bus **1702**. The storage unit **1716** and memory **1714** store the instructions **1710** embodying any one or more of the methodologies or functions described herein. The instructions **1710** may also reside, completely or partially, within the memory **1714**, within the storage unit **1716**, within at least one of the processors **1704** (e.g., within the processor's cache memory), or any suitable combination thereof, during execution thereof by the machine **1700**. Accordingly, the memory **1714**, the storage unit **1716**, and the memory of processors **1704** are examples of machine-readable media.

The I/O components **1718** may include a wide variety of components to receive input, provide output, produce output, transmit information, exchange information, capture measurements, and so on. The specific I/O components **1718** that are included in a particular machine **1700** will depend on the type of machine. For example, portable machines such as mobile phones will likely include a touch input device or other such input mechanisms, while a headless server machine will likely not include such a touch input device. It will be appreciated that the I/O components **1718** may include many other components that are not shown in FIG. **17**. The I/O components **1718** are grouped according to functionality merely for simplifying the following discussion and the grouping is in no way limiting. In various example embodiments, the I/O components **1718** may include output components **1726** and input components **1728**. The output components **1726** may include visual components (e.g., a display such as a plasma display panel (PDP), a light emitting diode (LED) display, a liquid crystal display (LCD), a projector, or a cathode ray tube (CRT)), acoustic components (e.g., speakers), haptic components (e.g., a vibratory motor, resistance mechanisms), other signal generators, and so forth. The input components **1728**

may include alphanumeric input components (e.g., a keyboard, a touch screen configured to receive alphanumeric input, a photo-optical keyboard, or other alphanumeric input components), point based input components (e.g., a mouse, a touchpad, a trackball, a joystick, a motion sensor, or other pointing instrument), tactile input components (e.g., a physical button, a touch screen that provides location and/or force of touches or touch gestures, or other tactile input components), audio input components (e.g., a microphone), and the like.

In further example embodiments, the components **1718** may include biometric components **1730**, motion components **1734**, environmental environment components **1736**, or position components **1738** among a wide array of other components. For example, the biometric components **1730** may include components to detect expressions (e.g., hand expressions, facial expressions, vocal expressions, body gestures, or eye tracking), measure biosignals (e.g., blood pressure, heart rate, body temperature, perspiration, or brain waves), identify a person (e.g., voice identification, retinal identification, facial identification, fingerprint identification, or electroencephalogram based identification), and the like. The motion components **1734** may include acceleration sensor components (e.g., accelerometer), gravitation sensor components, rotation sensor components (e.g., gyroscope), and so forth. The environment components **1736** may include, for example, illumination sensor components (e.g., photometer), temperature sensor components (e.g., one or more thermometer that detect ambient temperature), humidity sensor components, pressure sensor components (e.g., barometer), acoustic sensor components (e.g., one or more microphones that detect background noise), proximity sensor components (e.g., infrared sensors that detect nearby objects), gas sensors (e.g., gas detection sensors to detection concentrations of hazardous gases for safety or to measure pollutants in the atmosphere), or other components that may provide indications, measurements, or signals corresponding to a surrounding physical environment. The position components **1738** may include location sensor components (e.g., a Global Position system (GPS) receiver component), altitude sensor components (e.g., altimeters or barometers that detect air pressure from which altitude may be derived), orientation sensor components (e.g., magnetometers), and the like.

Communication may be implemented using a wide variety of technologies. The I/O components **1718** may include communication components **1740** operable to couple the machine **1700** to a network **1732** or devices **1720** via coupling **1722** and coupling **1724** respectively. For example, the communication components **1740** may include a network interface component or other suitable device to interface with the network **1732**. In further examples, communication components **1740** may include wired communication components, wireless communication components, cellular communication components, Near Field Communication (NFC) components, Bluetooth® components (e.g., Bluetooth® Low Energy), Wi-Fi® components, and other communication components to provide communication via other modalities. The devices **1720** may be another machine or any of a wide variety of peripheral devices (e.g., a peripheral device coupled via a Universal Serial Bus (USB)).

Moreover, the communication components **1740** may detect identifiers or include components operable to detect identifiers. For example, the communication components **1740** may include Radio Frequency Identification (RFID) tag reader components, NFC smart tag detection compo-

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nents, optical reader components (e.g., an optical sensor to detect one-dimensional bar codes such as Universal Product Code (UPC) bar code, multi-dimensional bar codes such as Quick Response (QR) code, Aztec code, Data Matrix, Data-glyph, MaxiCode, PDF417, Ultra Code, UCC RSS-2D bar code, and other optical codes), or acoustic detection components (e.g., microphones to identify tagged audio signals). In addition, a variety of information may be derived via the communication components 1740, such as, location via Internet Protocol (IP) geo-location, location via Wi-Fi® signal triangulation, location via detecting a NFC beacon signal that may indicate a particular location, and so forth.

What is claimed is:

1. A method of processing image data, the method comprising:
 receiving environmental data and associated capture time data from a sensor of a mobile computing device, the capture time data reflecting capture time of the environmental data, wherein the environmental data comprise audio data;
 processing the environmental data to identify, based on the audio data, a particular physical location at which the environmental data was captured, wherein the identifying of the particular physical location comprises:
 filtering the audio data using a low pass filter that excludes voice frequencies and isolates background audio data, thereby extracting non-voice environmental audio data;
 accessing stored audio fingerprint data that comprises multiple audio fingerprints associated with respective known locations;
 comparing the non-voice environmental audio data to the multiple audio fingerprints to identify the particular physical location, wherein predetermined first Global Positioning System (GPS) data is stored for the particular physical location in association with at least one of the multiple audio fingerprints; and
 correlating the predetermined first GPS data stored for the particular physical location in association with the at least one of the multiple audio fingerprints with second GPS data for a location as received by the mobile computing device;
 based on the processing of the environmental data, identifying in the audio data a verbal description, provided by a user of the mobile computing device during video capture, of subject matter contained in video data associated with the environmental data;
 in an automated operation, automatically generating a transcription of the verbal description;
 generating metadata based at least in part on:
 the particular physical location, and
 the transcription of the verbal description
 time stamping the metadata using the capture time data;
 receiving, at a processor, the video data and video time data associated with the environmental data, the video data comprising a plurality of video frames and the video time data reflecting record time of the video data;
 correlating the metadata to the video data using the capture time data and the video time data;
 receiving a search query, including a search criterion, at the processor;
 identifying a video frame within the video data by performing a search of the metadata using the search criterion; and
 including the identified video frame in a search result.

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2. The method of claim 1, wherein the processing of the environmental data further includes:

filtering the audio data using at least one of a band pass filter and a high pass filter, to extract voice-containing environmental audio data; and

performing voice recognition on the voice-containing environmental audio data to identify a word, and wherein the generating the metadata to include the word.

3. The method of claim 1, wherein the environmental data comprises location data, the method further comprising processing of the environmental data to identify at least one place, object or event using the location data, and wherein the generating the metadata includes the at least one place, object or event.

4. The method of claim 1, including performing image recognition with respect to the video frame, and tagging the video frame with a tag indicative of an object recognized within the video frame, the identifying of the video frame performed by matching the search criterion and the tag.

5. The method of claim 1, including performing characteristic recognition with respect to the video frame, and tagging the video frame with a tag indicative of a characteristic recognized in the video frame, the identifying of the video frame being performed by matching the search criterion and the tag.

6. The method of claim 1, comprising receiving segments of the video data by a messaging system as part of respective video messages, and combining the segments to constitute the video data.

7. The method of claim 6, wherein the respective video messages are ephemeral messages.

8. The method of claim 1, wherein the particular physical location comprises a museum exhibit inside of a museum building.

9. A system comprising:

at least one processor; and

at least one memory storing instructions that, when executed by the at least one processor, configure the system to:

receive environmental data and associated capture time data from a sensor of a mobile computing device, the capture time data reflecting capture time of the environmental data, wherein the environmental data comprise audio data;

process the environmental data to identify, based on the audio data, a particular physical location at which the environmental data was captured, wherein the identifying of the particular physical location comprises:

filtering the audio data using a low pass filter that excludes voice frequencies and isolates background audio data, thereby extracting non-voice environmental audio data;

accessing stored audio fingerprint data that comprises multiple audio fingerprints associated with respective known locations;

comparing the non-voice environmental audio data to the multiple audio fingerprints to identify the particular physical location wherein predetermined first Global Positioning System (GPS) data is stored for the particular physical location in association with at least one of the multiple audio fingerprints; and

correlating the predetermined first GPS data stored for the particular physical location in association with the at least one of the multiple audio fingerprints with a second GPS data for a location as received by the mobile computing device;

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based on the processing of the environmental data, identifying in the audio data a verbal description, provided by a user of the mobile computing device during video capture, of subject matter contained in video data associated with the environmental data;

in an automated operation, automatically generating a transcription of the verbal description;

generating metadata based at least in part on:

the particular physical location, and

the transcription of the verbal description

time stamping the metadata using the capture time data; receiving, at a processor, the video data and video time data associated with the environmental data, the video data comprising a plurality of video frames and the

video time data reflecting record time of the video data;

correlating the metadata to the video data using the capture time data and the video time data;

receiving a search query, including a search criterion, at the processor;

identifying a video frame within the video data by performing a search of the metadata using the search criterion; and

including the identified video frame in a search result.

10. The system of claim 9, wherein the identifying the particular physical location further comprises generating an audio fingerprint using at least a portion of the non-voice environmental audio data, and comparing the audio fingerprint to the multiple stored audio fingerprints.

11. The system of claim 9, wherein the identifying the particular physical location further comprises filtering the audio data to extract background audio data and the instructions, when executed by the at least one processor, configure the system to use the background audio data to identify the particular physical location.

12. The system of claim 9, wherein the instructions, when executed by the at least one processor, further configure the system to perform operations comprising:

filter the audio data using at least one of a band pass filter and a high pass filter, to extract voice-containing environmental audio data;

performing voice recognition on the voice-containing environmental audio data to identify a word included in the audio data; and

and to said generate the metadata to include the identified word.

13. The system of claim 9, wherein the environmental data comprises location data, and the instructions, when executed by the at least one processor, further configure the system to process the environmental data to identify at least one place, object or event using the location data, and to said generate the metadata to include the at least one place, object or event.

14. The system of claim 9, wherein the instructions, when executed by the at least one processor, further configure the system to perform characteristic recognition with respect to the frame, and to tag the frame with a tag indicative of a characteristic recognized in the frame, the identifying of the frame being performed by matching the search criterion and the tag.

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15. The system of claim 9, wherein:

the instructions, when executed by the at least one processor, further configure the system to receive segments of the video data by a messaging system as part of respective video messages, and to combine the segments to constitute the video data; and

the respective video messages are ephemeral messages.

16. A non-transitory computer-readable storage medium, the computer-readable storage medium including instructions that when executed by a computer, cause the computer to perform operations comprising:

receiving environmental data and associated capture time data from a sensor of a mobile computing device, the capture time data reflecting capture time of the environmental data, wherein the environmental data comprise audio data;

processing the environmental data to identify, based on the audio data, a particular physical location at which the environmental data was captured, wherein the identifying of the particular physical location comprises:

filtering the audio data using a low pass filter that excludes voice frequencies and isolates background audio data, thereby extracting non-voice environmental audio data;

accessing stored audio fingerprint data that comprises multiple audio fingerprints associated with respective known locations;

comparing the non-voice environmental audio data to the multiple audio fingerprints to identify the particular physical location, wherein predetermined first Global Positioning System (GPS) data is stored for the particular physical location in association with at least one of the multiple audio fingerprints; and

correlating the predetermined first GPS data stored for the particular physical location in association with the at least one of the multiple audio fingerprints with second GPS data for a location as received by the mobile computing device;

based on the processing of the environmental data, identifying in the audio data a verbal description, provided by a user of the mobile computing device during video capture, of subject matter contained in video data associated with the environmental data;

in an automated operation, automatically generating a transcription of the verbal description;

generating metadata based at least in part on:

the particular physical location, and

the transcription of the verbal description

time stamping the metadata using the capture time data; receiving, at a processor, the video data and video time data associated with the environmental data, the video data comprising a plurality of video frames and the

video time data reflecting record time of the video data;

correlating the metadata to the video data using the capture time data and the video time data;

receiving a search query, including a search criterion, at the processor;

identifying a video frame within the video data by performing a search of the metadata using the search criterion; and

including the identified video frame in a search result.

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